



TESTRX

USER MANUAL

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1. Introduction

1.1 Purpose

This document refers to the application by its abbreviated name, TESTRX, which stands for *Test Environment System for Tracking & Running Xecutions*.

2. About TESTRX

2.1 Overview

TESTRX is an integrated test environment designed for **test-case authoring**, **script generation** based on test bench configurations, and **test execution** for validating embedded automotive systems. The tool streamlines and standardizes the entire testing workflow through a unified and intuitive interface.

Key capabilities include:

- Authoring test cases in a standardized and reusable format
- Generating automated test scripts derived from test bench configuration parameters
- Supporting a wide range of test tools and environments, including **HIL systems** and **in-vehicle testing**
- Providing an intuitive graphical user interface for ease of use
- Ensuring standardization of all test artifacts, including test cases, test scripts, and results
- Offering a modular architecture that enables rapid customization and seamless integration with various tools in the testing ecosystem

3. TESTRX Installation and Uninstallation

- The application is designed to install and uninstall seamlessly on any Windows-based system or laptop.
- In the event of an installation error, the installer presents a clear notification and references the generated log file to facilitate troubleshooting.

3.1 Application Installation

- The application is packaged as an MSI installer to ensure a reliable, standardized, and easily manageable deployment process across different environments.
- This dual-mode installation approach enables IT administrators and end users to select the method best suited to organizational or individual requirements.

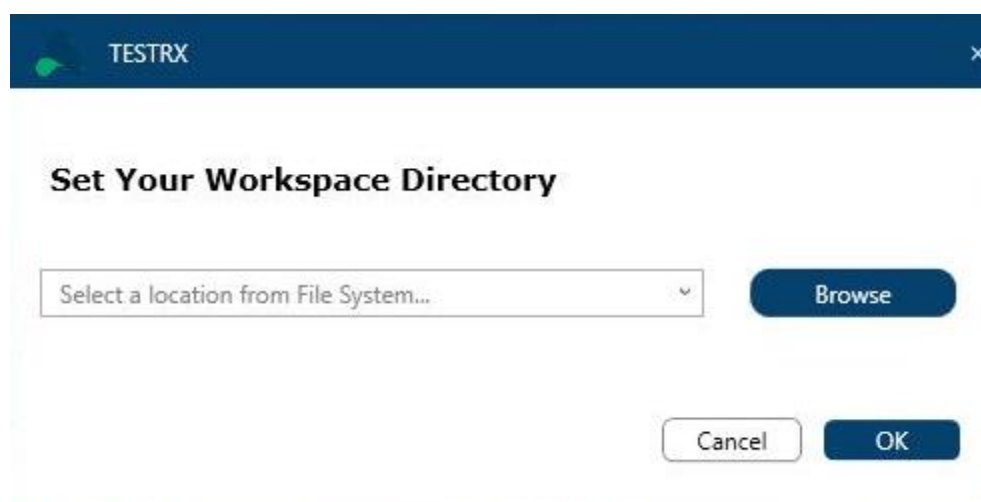
3.2 Application Uninstallation

- The application may be uninstalled through any of the following methods:
 - Control Panel → Programs and Features
 - Windows Settings → Apps & Features
- The uninstaller ensures system integrity by preserving shared dependencies required by other applications.
- In the event of an uninstallation failure, all changes are safely rolled back and the user is informed of the issue.

4. Workspace

4.1 Workspace selection on First Launch

- During the initial launch, the application automatically presents the **Workspace Selection** window, prompting the user to define the workspace location.
- Users may specify a workspace location either by manually entering a directory path or by selecting one through the Browse option.
- If the user selects “Cancel,” the operation is aborted and the application closes without storing any workspace information.



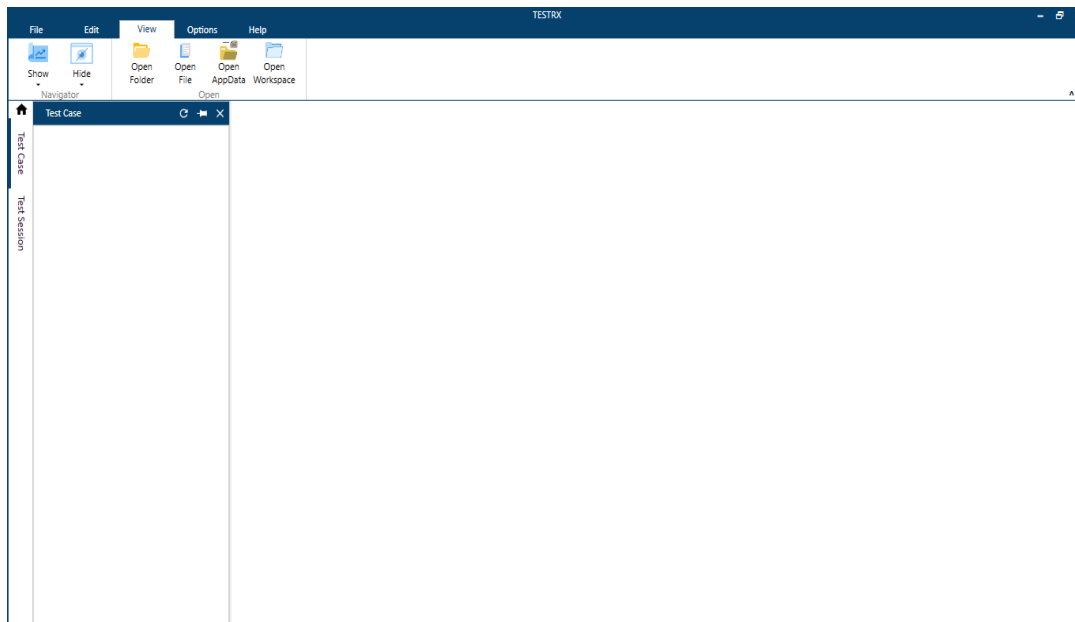
Snp.1 - Workspace selection

- Upon confirming the workspace location by selecting “**OK**,” the system generates a **TESTRX_Workspace** directory within the chosen path and automatically creates all required subfolders.

- After successful workspace configuration, the application proceeds to initialize essential components, load relevant data, and display the **Test Case** view.

5. Application Interface Overview

- The Home screen contains the Menu Bar, Test Environment Roles Selector , Project Navigator, and Test Case Editor.



Snp.2 - TESTRX Home screen

- The menu bar shows grouped items such as **File**, **Edit**, **View**, **Options** and **Help**. When clicked, each menu displays its related subcommands and actions.
- The menu bar can be collapsed using the **Up Arrow** button. It automatically restores and becomes visible again whenever any grouped menu item is clicked.
- The **Home tab** is the default landing page after launch and provides a quick view of project metrics, recent activity, and announcements.
- The **Test Case** tab lets users view, create, edit, and manage test cases within the selected project or test suite.
- The **Test Session** tab allows users to start, manage, and review test execution sessions.

6. Project Navigator

A hierarchical Project Navigator shall be displayed based on the selected user role:

- **Test Case Role:** Displays *Projects, Test Suites, and Test Cases*.
- **Test Session Role:** Displays *Test Sessions, Test Suites, and Test Cases*.

6.1 Navigator Panel Behaviour

- When the user clicks the Test Case or Test Session tab, the Project Navigator panel shall expand to occupy the left portion of the application window, providing access to the navigation tree.
- Clicking the same tab again shall collapse the Project Navigator panel and hide the tree view.

6.2 Hover and Overlay Behaviour

- When the user hovers over the Test Case or Test Session tab, the Project Navigator panel shall temporarily appear as an overlay, overlapping the current content (for example, the right panel).
- This behaviour provides quick access to project, test suite, and test case information without permanently occupying screen space.

6.3 Refresh, Pin and Close Controls

- When the Project Navigator is displayed as an overlay, a Refresh icon shall be available which on clicked will collapse the Project Navigator panel, hiding the tree view to maximize the working area while keeping the navigator tab accessible for quick re-expansion
- A **Pin icon** shall be available within the navigator panel, clicking the Pin icon shall lock the Project Navigator in an expanded state.
- Once pinned, the navigator shall remain visible until the user manually collapses or unpins it.
- A **Close icon** is available on the navigator panel, allowing the user to hide the Project Navigator panel from the main window.

6.4 Node Interaction

- On **double-clicking** a selected node (for example, a Test Case or Test Suite) within the Project Navigator, a **new tab** shall open in the right panel.
- The tab shall display detailed information or the appropriate editor view related to the selected node.

7. Menu Bar

The application provides a **menu bar at the top of the window**, enabling users to easily and consistently access the available features and functionalities. The menu bar is organized into clearly defined menus, each grouping related actions for efficient navigation and usability.

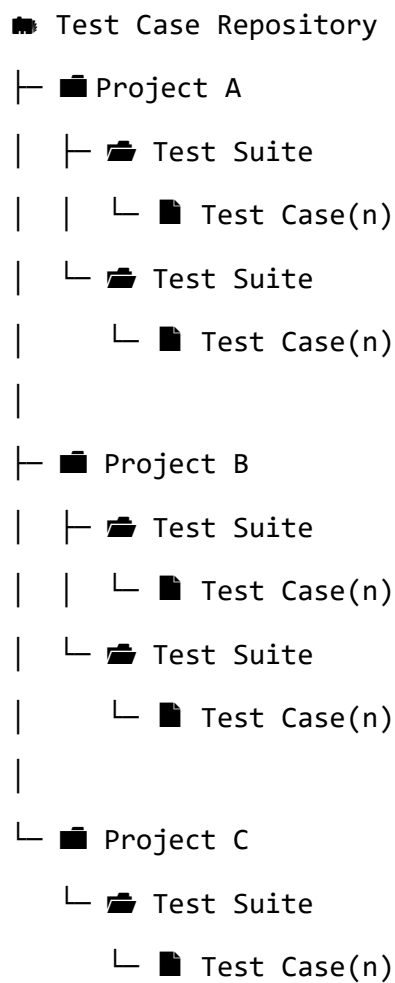
7.1 Menu and Sub-Menu Description

Menu	Sub Menu			Description
File	Save			Saves the currently active test case changes.
	Save All			Saves all open test case tabs pending changes.
	New	Project	Load	Load project from ALM
			Create New	Create New Project
		Suite		Creates a new test suite within the project.
		Case	Heading	Creates a new heading to organize test cases.
			System Test	Creates a new test case.
			Test - System	Creates test case that can be flushed to ALM
		Open		
	Close			Closes a Project or Test Suite
	Import			Imports an exported Test Suite or Test Case
	Export			Exports the selected Test Suite or Test Case
	Excel			Export the selected Test Suite or Test Case to Excel
	Workspace	Add Workspace		Create a new workspace
		Clean Up		
	Flush			To flush a Test Suite or Test Case to ALM
Edit	Cut			Cut the selected Test Suite or Test Case
	Copy			Copy the selected Test Suite or Test Case
	Paste			Paste the copied Test Suite or Test Case
	Rename			Rename selected Test Suite or Test Case or Project
	Delete			Delete selected Test Suite or Test Case or Project
	Recover			Future Scope
	Optimize			Future Scope
View	Show	Test Case	Displays the Test Case tab	
		Test Session	Displays the Test Session tab	
	Hide	Test Case	Hides the Test Case tab	
		Test Session	Hides the Test Session tab	

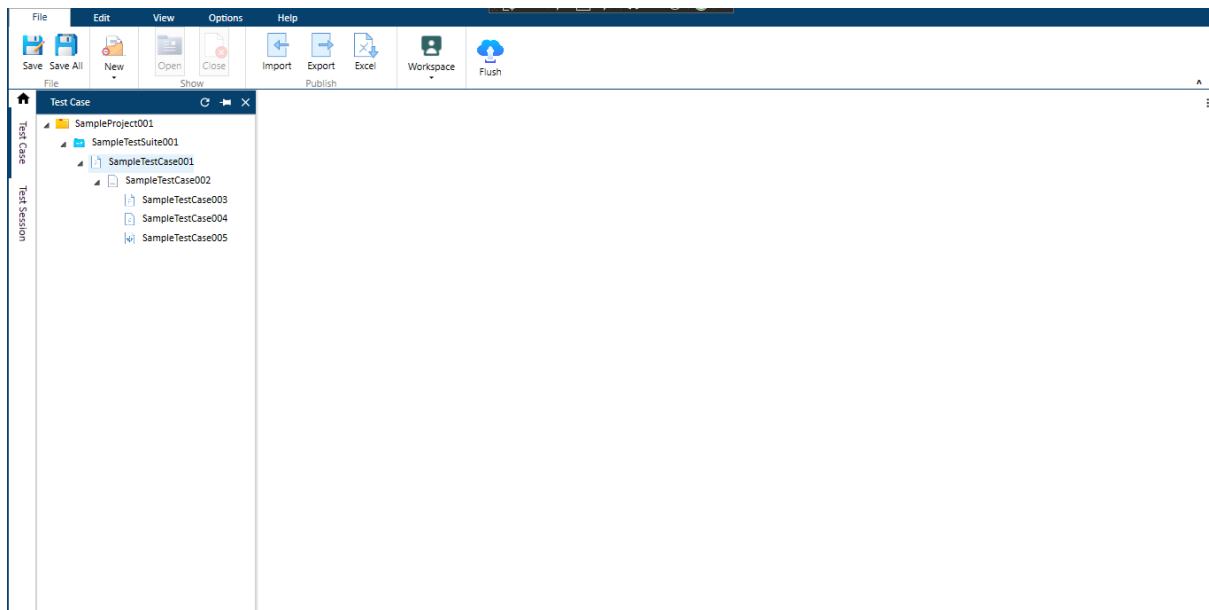
	Open Folder	Opens the associated folder location.
	Open File	Opens .json file for the selected node
	Open App Data	Opens App Data folder
	Open Workspace	Opens the current workspace location.
Options	Cleanup	Future Scope
	User Preferences	Future Scope
	Logs	Future Scope
Help	User Manual	Opens User Manual in browser
	About	Future Scope
	Download User Manual	Future Scope
	Download FAQs	Future Scope
	FAQ	Future Scope
	Support	Future Scope

Table 1 : Menu Bar options

8. Test Case Hierarchy



Snp.3 – Test Case Repository Structure



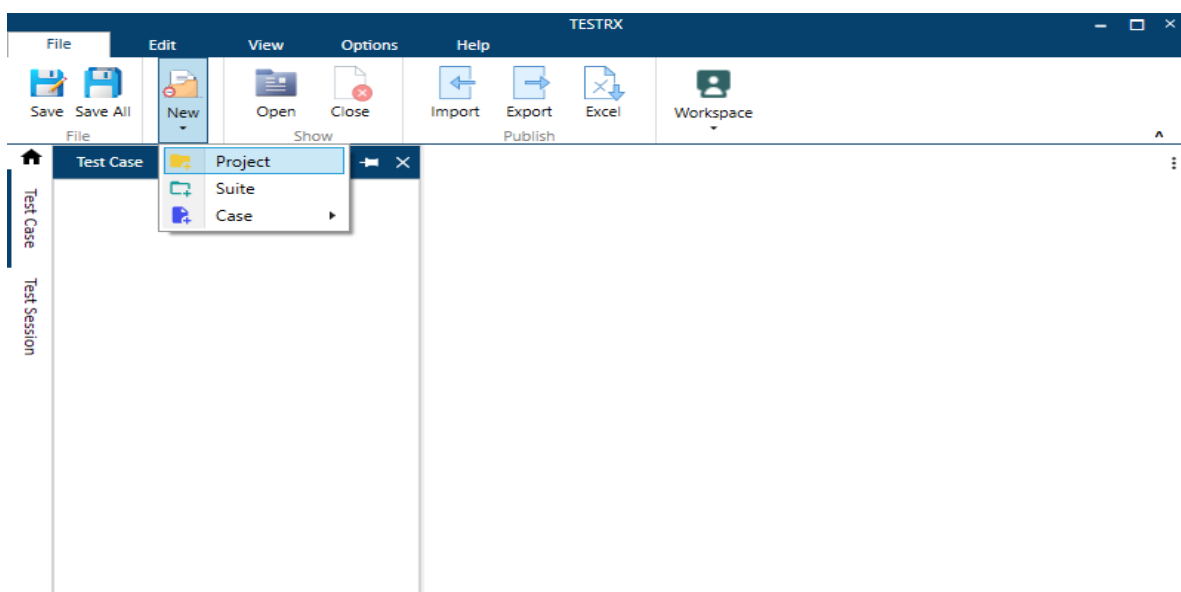
Snp.3 – Test Case Hierarchy

9. Project Creation

Before authoring test cases in TESTRX, a project must be created within the application. Users can initiate project creation through the **Create New Project** workflow.

Procedure to add a project:

- Open the File menu and select New.
- From the submenu, choose Project to begin the project creation process.



Snp.4 – Add a new project

- Project folder right-click options

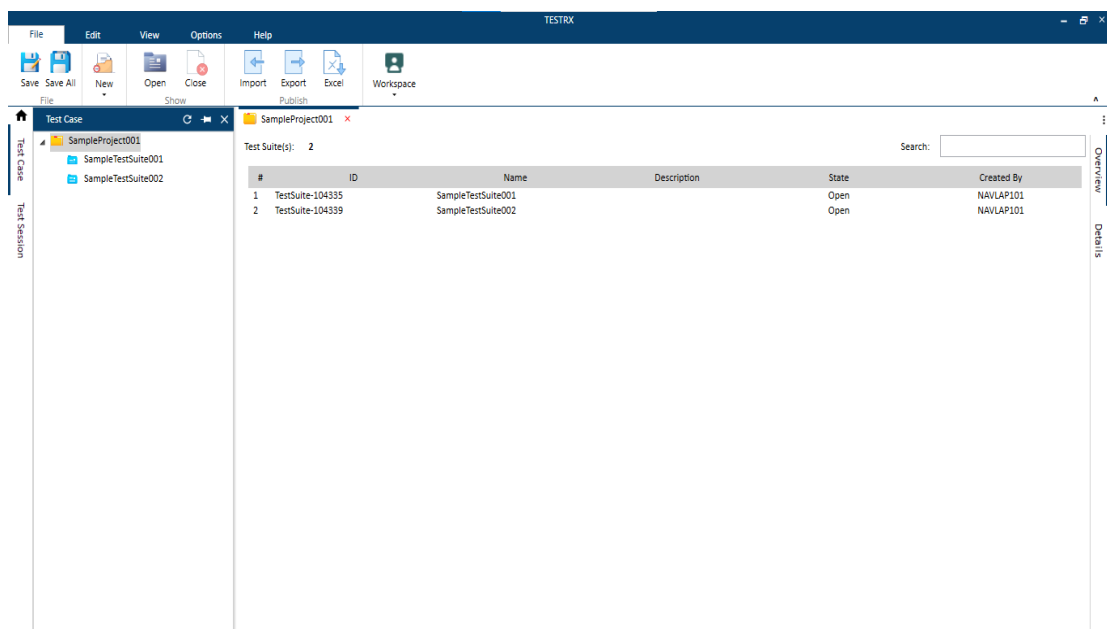
Selected Item	Context Menu Options	Actions Performed
Project – Navigator Window	Add Test Suite	Creates a new test suite.
	Import Test Suite	Imports an existing test suite file.
	Paste Test Suite	Pastes a copied test suite into the selected location.
	Rename	Renames the selected item.
	Recover Test Suite	Restores a closed test suite.
	Close Project	Closes the current project.
	Delete Project	Deletes the selected project.

Table 2: Right click options on Project

9.1 Overview and Details

9.1.1 Overview Tab

- Double-clicking a project or test suite displays its corresponding overview in the Overview panel.



Snp.5 – Project Overview

- The system displays detailed information for all Test Suites within the selected project and all Test Cases within the selected Test Suite. The overview includes the following attributes:

- **ID** – A system-generated unique identifier assigned to each Test Suite or Test Case.
- **Description** – A brief summary outlining the purpose and scope of the Test Suite or Test Case.
- **State (Open/Closed)** – Indicates whether the item is currently open for editing or closed.
- **Created By** – Specifies the user who originally created the Test Suite or Test Case.
- **Category** – Specifies the classification of the Test Case, such as whether it is a System Test or a Heading.

9.1.2 Details Tab

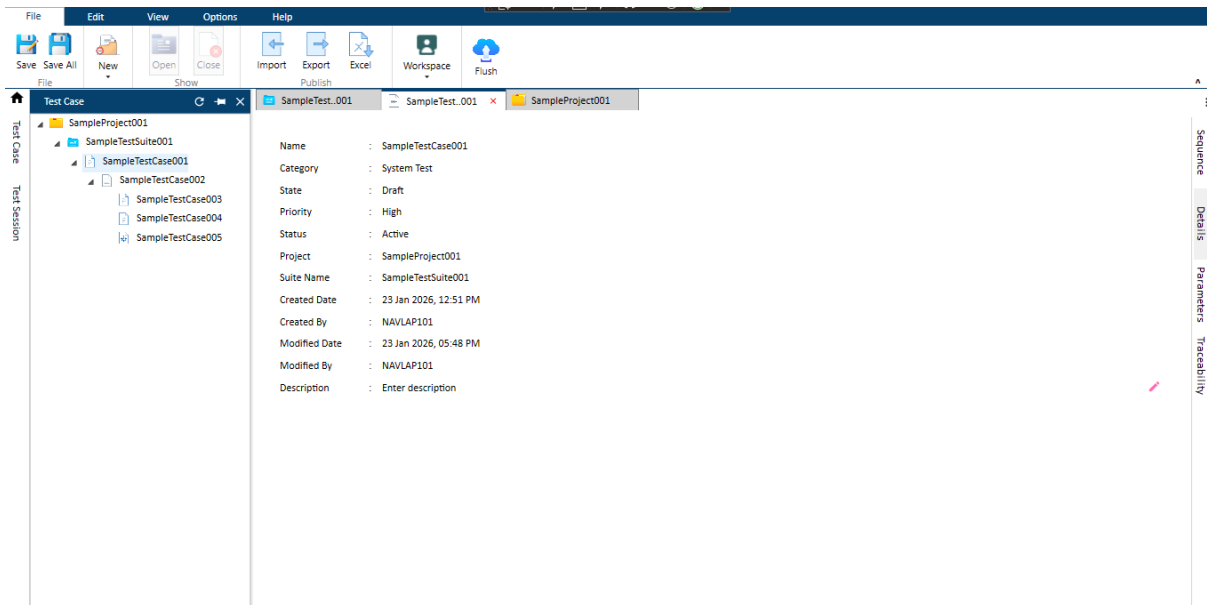
- The Details option is available at every level and provides comprehensive information. The detailed information can be viewed by clicking the Details tab, as described below.

Item	Operation performed
Project	Double click on Project – select Details tab
Test Suite	Double click on Test Suite – select Details tab
Test Case	Double click on Test Case – select Details tab
Heading	Double click on Heading – select Details tab

Table 3: View More Details

- The Details tab presents information for all Projects, Test Suites, Test Cases, and Headings. It includes the following key attributes:
 - **Name** - Specifies the title assigned to the Project, Test Suite, Test Case, or Heading.
 - **State** - Indicates the current editing status of the item, such as Open or Closed.
 - **Created Date** - Displays the date and time when the item was originally created.
 - **Created By** - Identifies the user who created the item.
 - **Modified Date** - Shows the date and time when the item was last updated.
 - **Modified By** - Specifies the user who performed the most recent modification.
 - **Description** - Provides a brief summary explaining the purpose, scope, or functionality of the item.
 - **Project** - Indicates the project under which the Test Suite, Test Case, or Heading is organized.

- **Short Name** - Represents a concise identifier used for easier recognition.
- **Category** - Defines the classification of the item, such as System Test, Heading.
- **Priority** - Indicates the importance level assigned to a Test Case, typically used for planning and execution order.
- **Suite Name** - Identifies the Test Suite to which the Test Case or Heading belongs.



Snp.6 - Details tab of Test Case

10. Create Test Suite

A Test Suite can be created in TESTRX only after a project has been added. TESTRX provides two methods to create a new Test Suite:

- **Method 1: Using the Context Menu**
 - 1 Right-click the added project in the Project Navigator.
 - 2 Select Add Test Suite from the context menu.
- **Method 2: Using the File Menu**
 - 1 Select the project from the Project Navigator.
 - 2 Navigate to the File menu and click the New submenu.
 - 3 Choose Suite to create a new Test Suite.

10.1 Test Suite Sub Tabs

The following sub-tabs are available for Test Suites:

- **Overview** – Displays all Test Cases contained within the Test Suite.
- **Details** – Presents the configuration and descriptive information of the Test Suite.

- **Signia** – Allows users to add database files such as A2L, DBC, and LDF.
- **Parameters** – Lists all user-defined parameters utilized during Test Case authoring in the TESTRX editor.
- **Labels** – Displays database parameters and user-defined labels applied while authoring Test Cases in the TESTRX editor.

10.2 Test Suite Context Menu Options

The following options are available when right-clicking a Test Suite:

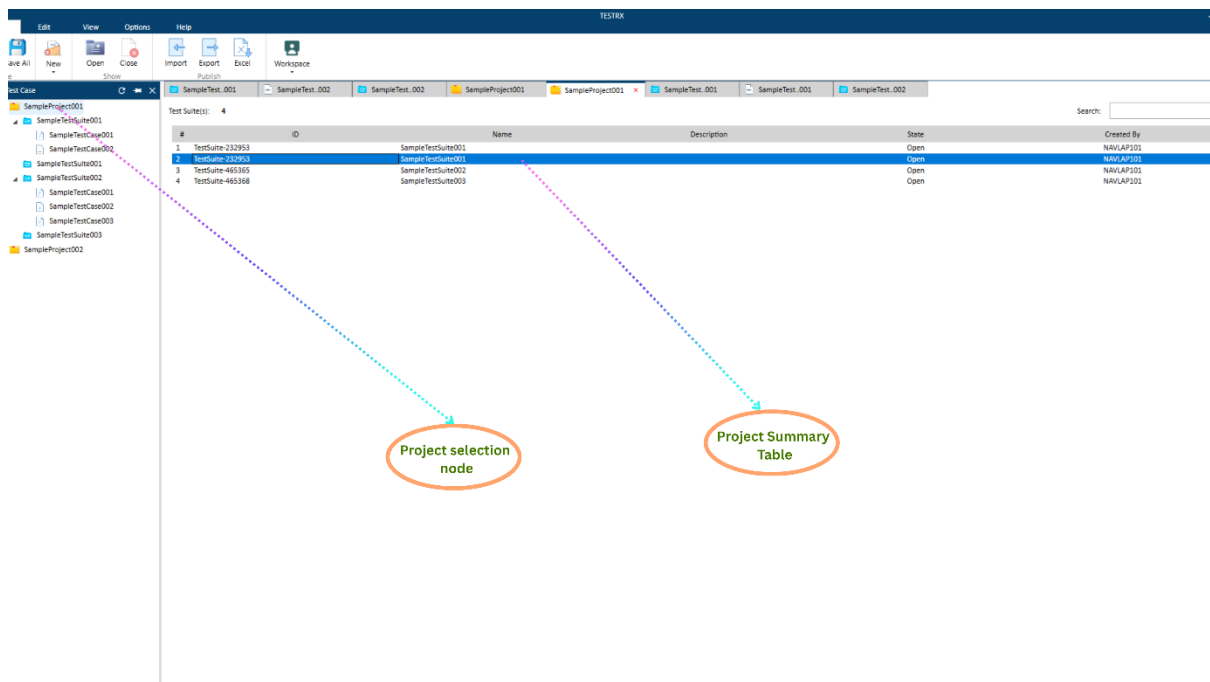
Selected Item	Context Menu Options	Action Performed
Test Suite	Add Test Case	For creating a new Test Case
	Import Test Case	The selected Test Suite will imported from Zip file
	Paste Test Case	To Paste the Copied Test Case
	Rename	To Rename the Test-Suite
	Copy Test Suite	To copy the Test Suite
	Export Test Suite	The selected Test Suite will exported as a .Zip file
	Export to Excel	Export the selected Test Suite to Excel
	Close Test Suite	To close the selected Test Suite
	Delete Test Suite	To delete the selected Test Suite

Table 4: Right click options on “Test Suite”

10.3 Accessing a Test Suite from the Project Summary Table

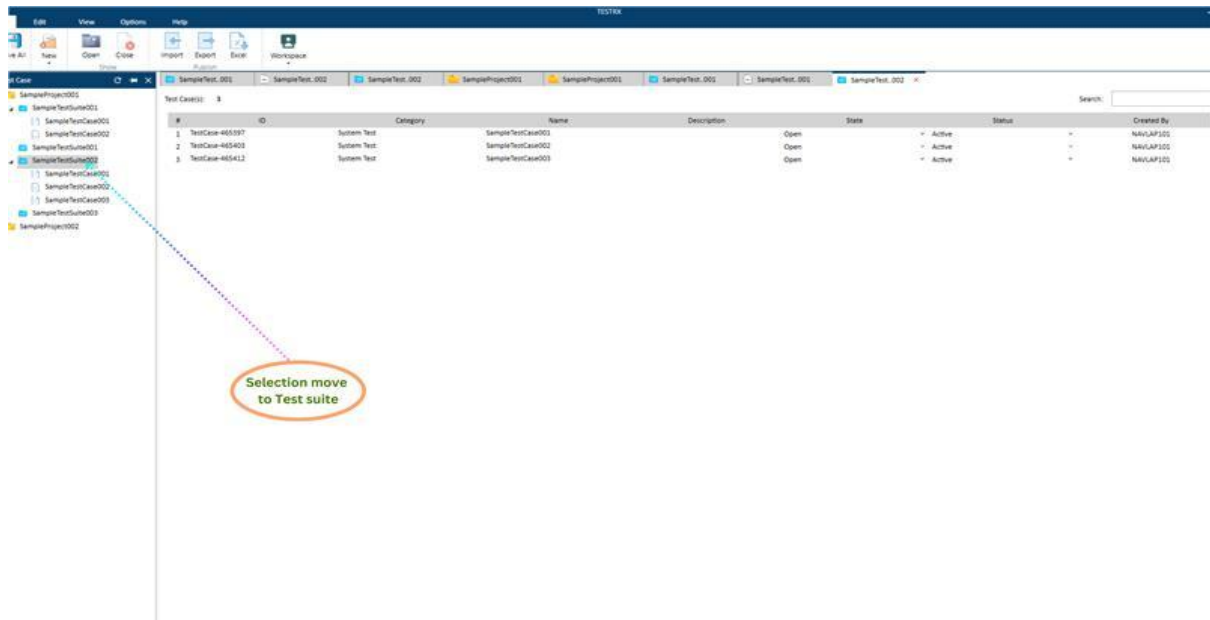
A Test Suite can be opened by double-clicking its entry in the Project Summary table.

Double click on the test suite:



Snp.7 – Double clicking on the Test Suite

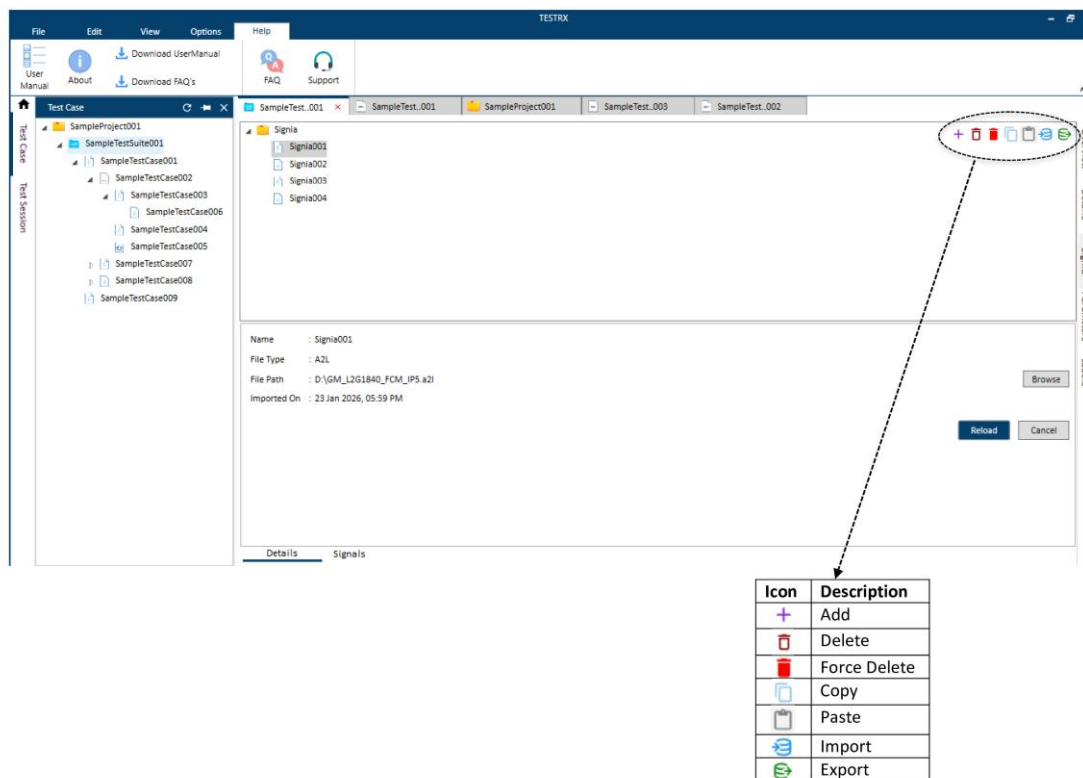
After double clicking selection will move to test suite



Snp.8 – Selection moved to Test Suit

11. Signia

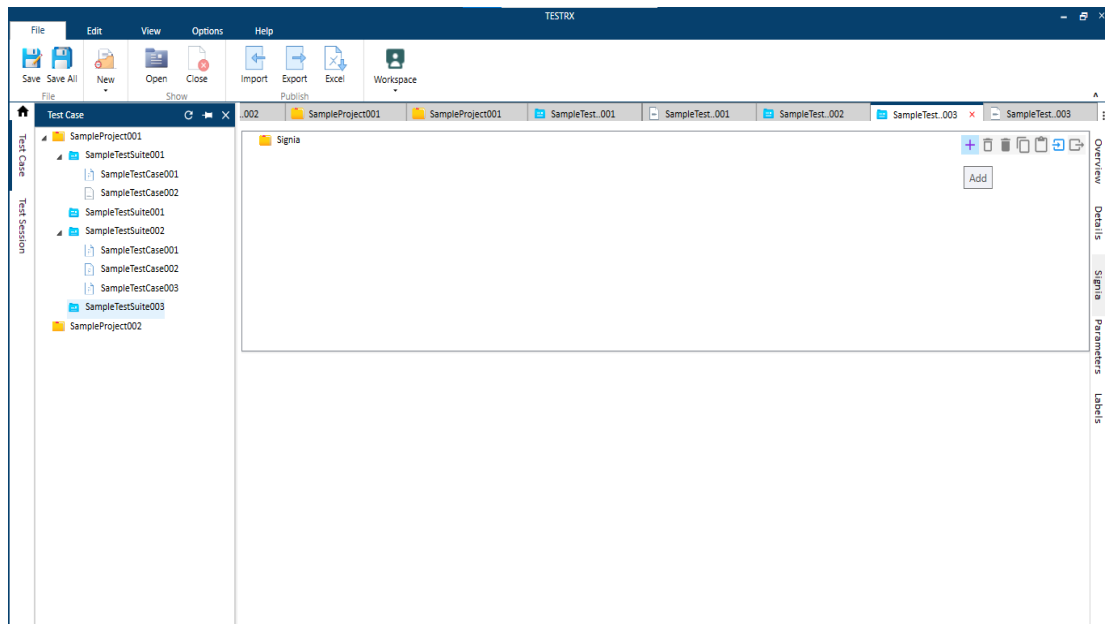
- The Signia tab facilitates the management of database files (such as A2L, DBC, and LDF) associated with a Test Suite.
- The interface is structured into two sections: an upper panel that displays the list of imported files, and a lower panel that presents detailed information for the selected file.
- The lower panel contains two sub-tabs: **Details** and **Signals**.
 - The **Details** sub-tab displays key information about the selected database file, including its name, file type, file path, and import timestamp.
 - The **Signals** sub-tab presents a list of all signals defined in the database file, along with their attributes such as label name, data type, conversion, and minimum and maximum values.



Snp.9 – Signia Tab

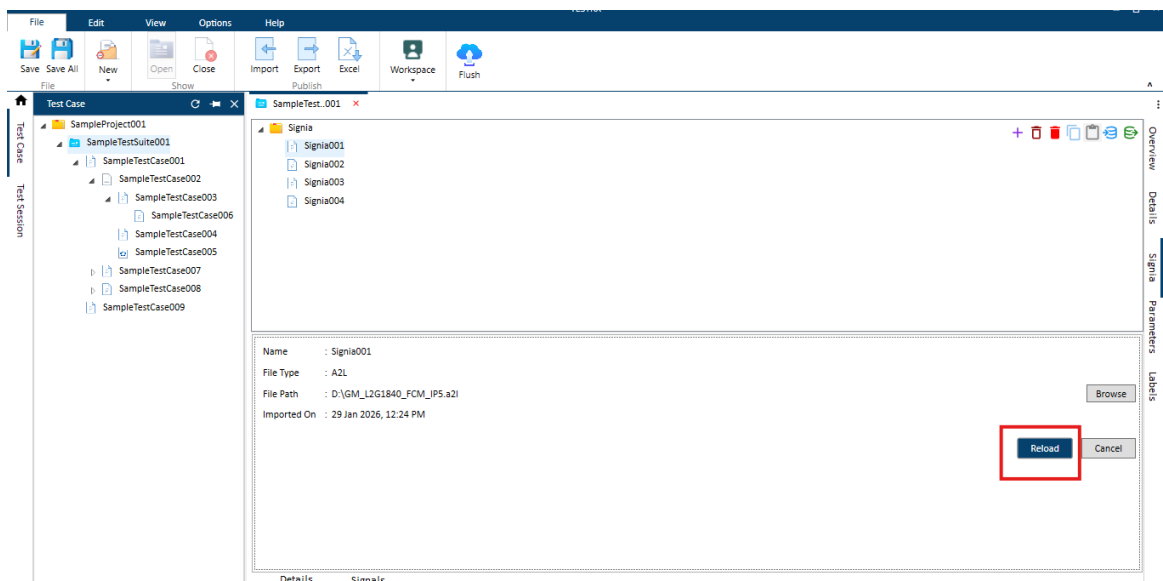
11.1 Adding Database

- To add a database, click the Add(+) Icon in the upper panel of the Signia tab and select the required file type (A2L, DBC, or LDF).



Snp.10 – Add Function for Label Database Import

- In the lower panel, use the Browse option in the Details section to locate and select the desired file, then click Save to confirm the import.
- Upon saving, a confirmation message will appear indicating successful import. After acknowledging the message, the Imported On field will display the date and time of the import.
- **Reload** option can be used to update the existing database by browsing and selecting a new or modified database file; clicking **Reload** re-imports the selected file and refreshes all data with the latest content.



Snp.11 – Reload Database File

11.2 Delete and Force Delete – Database File

- These options are available in the upper panel of the Signia tab.
- Allows removal of a selected database file from the Signia configuration.
- The **Delete** option performs dependency checks and prevents removal if the file is referenced by active Test Cases.
- The **Force Delete** option bypasses dependency validation and removes the file immediately.
- The original file on the system remains unchanged; only its reference in the application is removed.
- Force Delete should be used cautiously, as it overrides standard safeguards and may require updates to dependent Test Cases.

11.3 Copy and Paste – Database File

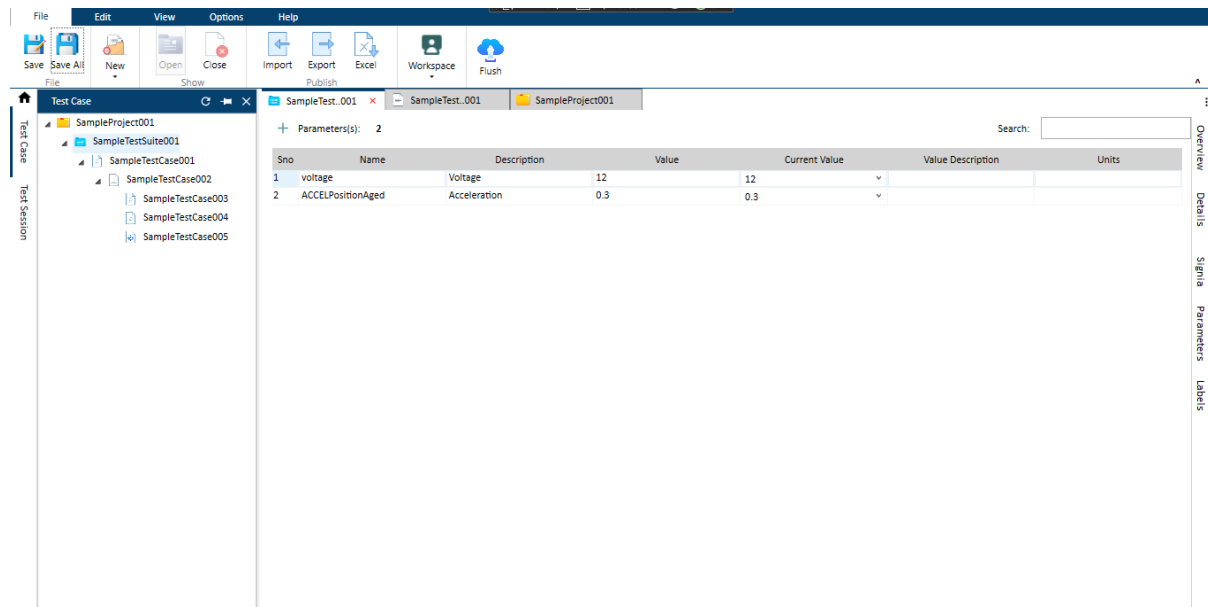
- These options are available in the upper panel of the Signia tab.
- The **Copy** option allows the selected database file to be copied from its current Test Suite.
- The **Paste** option enables the copied database to be inserted into another Test Suite.
- Only the reference within the application is duplicated; the original file on the system remains unchanged.
- If the same database file is copied and attempted to be pasted into the same Test Suite, the tool displays an error message indicating that the file already exists.

11.4 Import and Export – Database File

- The Import and Export options are available in the upper panel of the Signia tab.
- The **Import** option allows users to add a database file (A2L, DBC, LDF) to the Test Suite that was exported from another test suite.
- Imported files appear in the upper panel, and their details are shown in the lower panel.
- If an invalid file is selected, the system displays an error message.
- The **Export** option enables users to save the selected database file to a chosen directory.
- The exported file retains all its signal information for reuse in other Test Suites.
- If the export operation fails, an error popup is displayed.

12 Parameters

The **Parameters** feature allows users to define, manage, and apply Test Suite–level and Test Case–level parameters. Parameters support dynamic value substitution within Test Cases using the format `{{parameter_name}}`.



Snip.12 – Parameters Tab

12.1 Creating Parameters

- Parameters can be created from the **Parameters** tab at both the Test Suite and Test Case levels, or directly within the **Editor**.
- In the Editor, parameters are defined using the syntax `{{parameter_name}}`.
- In the Parameters tab, new parameters can be added by selecting the '+' option.
- When a parameter is created through the Parameters tab, the system assigns a default name in sequential order (e.g., *Parameter001*, *Parameter002*, etc.).
- Once created, the parameter row enters editable mode, allowing the user to rename and configure it immediately.
- Pressing **Enter** confirms the parameter name and inserts it into the parameter table.
- Parameters remain local to their context (Test Suite or Test Case) until explicitly saved.
- Parameters can be saved by Save option in the Menu Bar or by using shortcut "Ctrl+S"

12.2 Parameter Types and Scope

12.2.1 Test Suite–Level Parameters

- These parameters are global in scope and accessible across all Test Cases contained within the Test Suite.
- Test Suite parameters provide default values that may be selectively overridden at the Test Case level.

12.2.2 Test Case–Level Parameters

- These parameters are local to the Test Case in which they are defined.
- Test Case parameters take precedence over Test Suite parameters when both share the same name.
- Newly created Test Case parameters remain visible only within the specific Test Case until they are saved.

12.2.3 User-Defined Parameters

- Parameters added through the editor are automatically captured and listed in the parameter table once the user confirms the input.
- Multiple parameters may be entered within a single editor cell.

12.3 Parameter Naming Rules

Parameter names must adhere to strict validation rules to ensure consistency, clarity, and compatibility:

- Must begin with an alphabetic character (A–Z or a–z).
- May contain only alphanumeric characters and underscores (_).
- Must not contain spaces.
- Must be longer than three characters.
- Must not begin or end with an underscore.
- Must not contain consecutive underscores.
- Must not be identical to reserved keywords in the target programming language.

- Must not consist solely of numeric characters or numeric groups separated by underscores.

12.4 Parameter Table Structure

Each parameter includes the following fields for configuration and reference:

- **Name** – The unique identifier of the parameter.
- **Description** – The purpose or functional meaning of the parameter.
- **Value** – The predefined list of acceptable values.
- **Current Value** – The value currently assigned for use during execution.
- **Value Description** – A detailed explanation of the selected value.
- **Units** – Measurement units associated with the parameter value.
- **Set In** – Indicates whether the parameter belongs to the Test Suite or Test Case.

12.5 Parameter Value Rules

12.5.1 Value Field

The **Value** column supports:

- A single numeric value
- A comma-separated list of numeric values

Validation Rules:

- Values must be numeric.
- Duplicate values are not permitted.
- Entries must be separated by commas.
- No trailing or consecutive commas allowed.

Invalid Example:

1,,2 or 1,2,

Valid Example:

1,2,3

12.5.2 Current Value Field

The user may:

- Select a value from the predefined list, or
- Enter a new value manually, provided it satisfies validation rules.

Additional behaviors:

- Newly entered valid values are automatically added to the **Value** list.
- When switching between values, the corresponding **Value Description** and **Units** are preserved and reloaded.
- Edits to value descriptions and units apply only to the selected value.

12.6 Renaming Parameter

- Parameter names can be edited only at the Test Suite level.
- If the parameter is not referenced in any Test Case, renaming occurs immediately.
- If the parameter is used in one or more Test Cases, a warning dialog appears:

Parameter Warning

“The selected parameter is already used in Test Case!”

To rename from old_param to new_param, click Continue.

To add it as a new parameter, click Create New.”

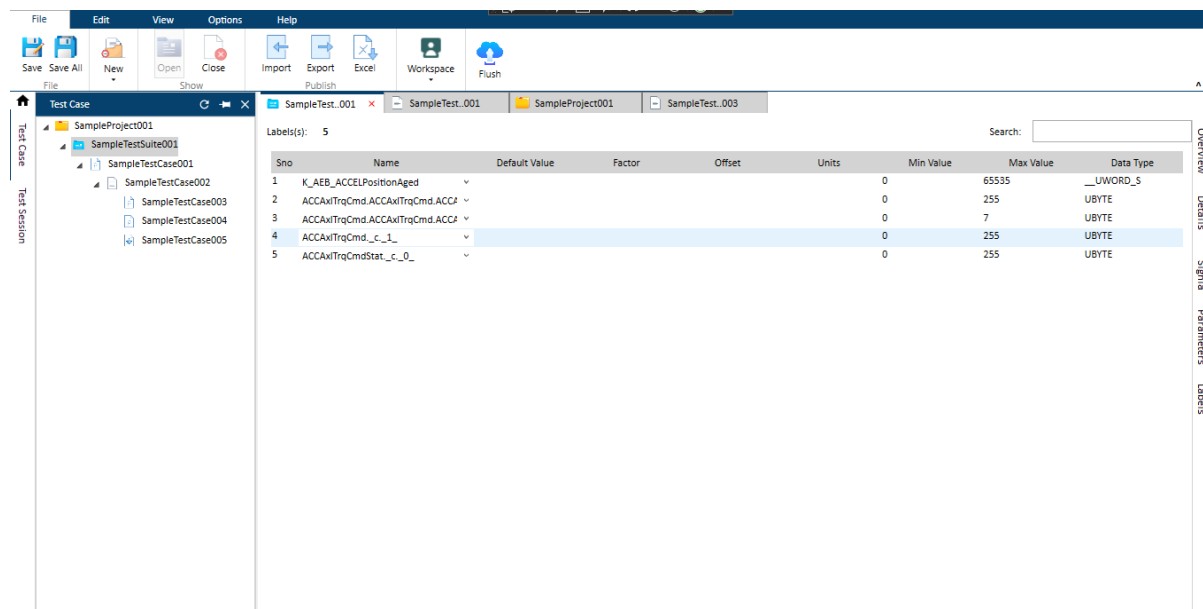
- All naming rules are revalidated before finalizing the rename.

12.7 Removing Parameters

- A right-click on a parameter row in the parameter tab opens the context menu with **Remove Parameter**.
- Removal is confirmed via popup:
Confirm deletion of the parameter “parameter_name”?
- Deletion occurs only after user confirmation.

13 Labels

The **Labels** tab is displayed on the right side of the interface, similar to the **Details** tab, and is available exclusively at the **Test Suite** level. This tab provides a consolidated view of all labels referenced across the Test Cases within the current Test Suite.



Snp.13 – Labels Tab

13.1 Label Availability and Creation

- The Labels tab lists all labels used in any Test Case under the selected Test Suite.
- Users may create new labels within the Test Case editor by entering them in appropriate column *Identifier*.
- Newly created labels become available in the Labels tab only after the user **saves** the changes.
- Saved labels are made globally accessible across all Test Suites, ensuring consistency and centralized management.

13.2 Labels Tab Columns

The Labels tab displays the following columns:

- **Name** – Displays the label name referenced within the Test Cases. This is the only field that can be modified by the user.
- **Default Value** – Shows the default value fetched from the associated database file (A2L, DBC, LDF). This value is system-derived and not user-editable.

- **Factor** – Represents the scaling factor applied to the raw signal value; automatically populated from the imported metadata.
- **Offset** – Indicates the offset used in signal conversion calculations; auto-filled based on the label's source definition.
- **Units** – Specifies the measurement units associated with the label (e.g., °C, km/h), derived automatically from metadata.
- **Min Value** – Displays the minimum permissible value for the label as defined in the source file.
- **Max Value** – Displays the maximum permissible value for the label as defined in the source file.
- **Data Type** – Indicates the data type of the label (e.g., uint8, float, int16), determined from the imported database.
- **Description** – Provides a brief explanation or meaning of the label, populated from available metadata in the source file.

13.3 Label Management Options

13.3.1 Remove Label

- Users can remove labels directly from the Labels tab.
- If a label is referenced in any Test Case, the system prevents removal and displays a warning:
"Cannot remove label as it is used in Test Case(s)."

13.3.2 Renaming Labels

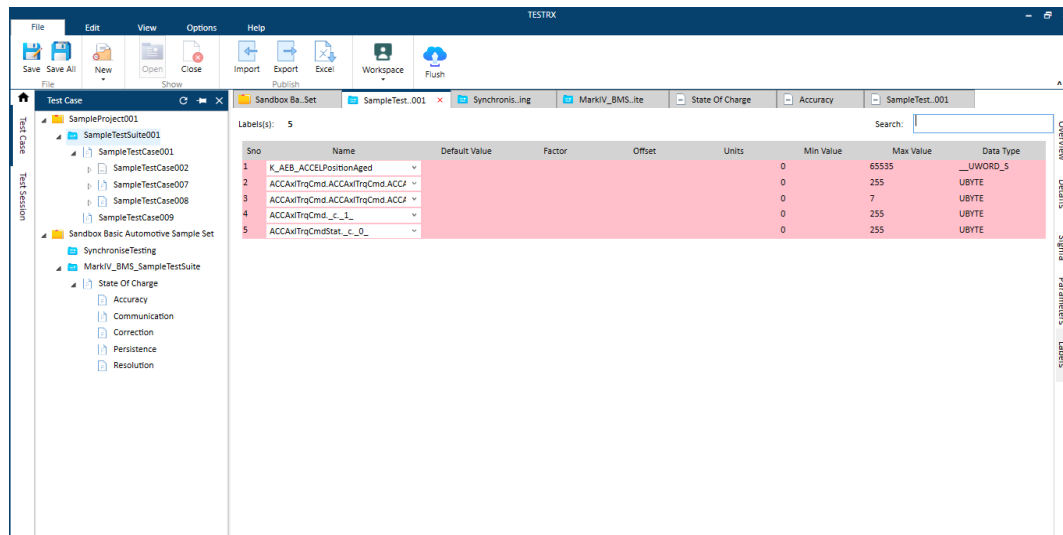
When a user attempts to rename a label:

- If the new name already exists within the same Test Case or Test Suite, the system shows a confirmation popup:
"A label with the name '<new_name>' already exists. Do you want to merge the current label with the existing one?"
- **Yes** → The system merges all associated reference data into the existing label.
- **Cancel** → The rename is aborted with no changes applied.

13.4 Visual Indicators for Label Validity

- The system applies a pink background to a label row under the following conditions:

- **Signia file deletion:** Linked labels are removed or references become invalid.
- **Signia file reload:** Updated Signia file removes labels previously available.
- **Test Case copy:** A copied Test Case references labels without a corresponding Signia file in the new suite.
- **Step copy:** Same condition as above for step-level copying.



Snp.14 – Pink Labels

- The UI updates the row's colour immediately whenever a relevant Signia change occurs.

14 Create Test Case

- A Test Case can be created in two ways:
 1. Right-click the desired Test Suite or Test Case and select Add Test Case or Add Heading.
 2. Use the New menu, which provides two options—Test Case and Heading—and select the desired option to create.
- A new node is inserted into the selected Test Suite/Test Case and opened in inline edit mode.
- Default names follow incremental formats:

Test Case: TestCase001, TestCase002, ...

Heading: Heading001, Heading002, ...

14.1 Naming Rules

- A Test Case name may include letters, numbers, spaces, and the underscore (_).
- The system enforces the following:
 - Name cannot be empty → “Case name cannot be empty.”
 - Maximum length: 150 characters
 - Cannot begin with a special character
 - Leading/trailing spaces are trimmed automatically
 - Invalid characters are not allowed: , : * ? " < > | @ # \$ etc.
 - Reserved terms may be restricted (e.g., default, null, system)
→ “The name uses a reserved keyword. Please choose a different name.”
 - Pressing Enter confirms the name and saves the Test Case.

14.2 Test Case Context Options

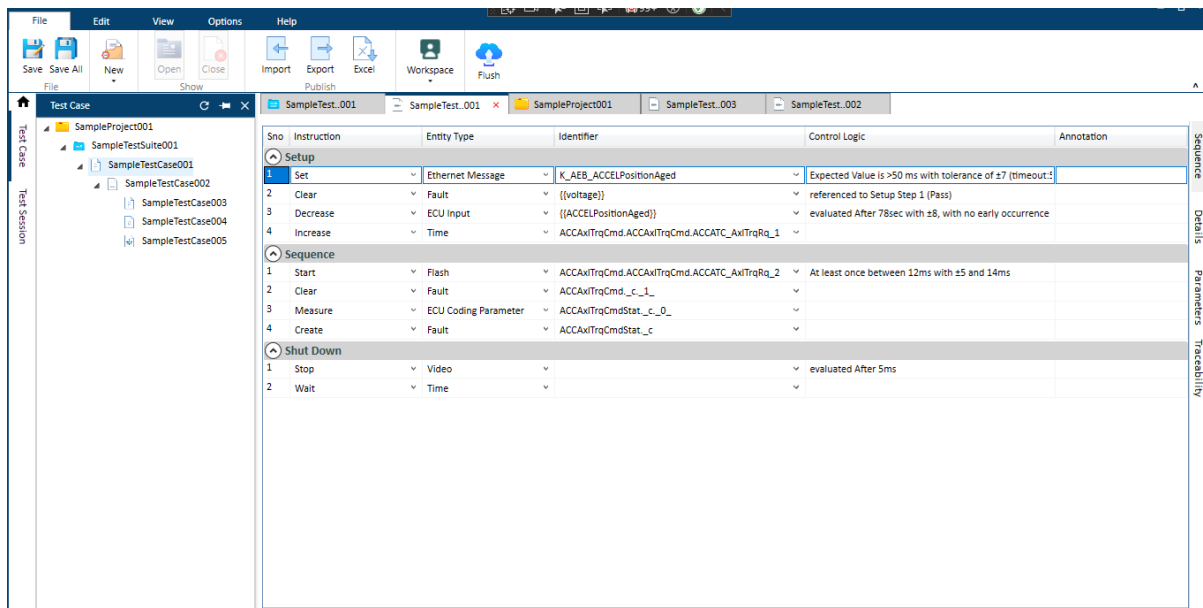
The following options are available when right-clicking a Test Case:

Selected Item	Context Menu Options	Action Performed
Test Case	Add Test Case	For creating a new Test Case
	Import Test Case	The selected Test Suite will imported from Zip file
	Paste Test Case	To Paste the Copied Test Case
	Rename	To Rename the Test-Case
	Copy Test Case(s)	To copy the Test Case
	Export Test Case	The selected Test Case will exported as a .Zip file
	Export to Excel	Export the selected Test Case to Excel
	Delete Test Suite	To delete the selected Test Suite

Table 5: Right click options on “Test Case”

14.3 Test Case Editor – Sequence Tab

- The **Sequence** tab becomes available when a test case is selected and is displayed in the right-side panel. This tab allows users to **edit the test case**, including modifying steps and configurations.



Snp.15 – Test Case Editor

- The Sequence tab displays a fixed-layout table organized into three distinct sections, each identified by a labelled header:
 - Setup:** Defines the automated preconditions and initialization steps required before test execution, such as environment configuration, data preparation, and system setup.
 - Sequence:** Contains the automated test steps that drive the core execution logic, including actions, validations, and control flow.
 - Shut Down:** Specifies automated clean-up activities and post-execution validations, such as resource release, log collection, and system reset.
- The following columns are available for defining each test step:
 - Instruction:** Defines the automated action to be executed as part of the test step (e.g., Set, Check, While).
 - Entity Type:** Specifies the type of component the instruction acts upon (e.g., CAN Signal, Ethernet Message).
 - Identifier:** Represents the user defined label or parameter used to configure the entity
 - Control Logic:** Defines the conditional logic governing the execution of the step (e.g., Expected Value is >20 with tolerance of 1%).

- **Annotation:** Allows users to add comments or descriptive notes for documentation, clarity, or maintenance purposes (e.g., Validates successful login, Precondition: Application is launched).

14.4 Test Step Context Menu

The Test Case Editor provides a context menu to efficiently manage test steps. This menu is accessible by right-clicking on a “Sno” column within the Sequence tab.

Option	Description
Insert Step	Inserts a new test step below the selected step.
Insert Nested Step	Inserts a nested step under the selected step.
Undo	Reverts the most recent action performed in the editor.
Redo	Reapplies the last undone action.
Clear Step(s)	Clears the content of the selected step(s) without deleting them.
Remove Step(s)	Deletes the selected step(s) from the test case.
Convert Step ↔ Nested Step	Converts a step to a nested step or a nested step to a normal step.
Copy Step(s)	Copies the selected step(s) to the clipboard.
Cut Steps(s)	Cuts the selected step(s) and places them on the clipboard.
Paste Step(s)	Pastes the copied or cut step(s) at the selected location.

Table 6: Test Step Context Menu Options

14.5 Key-Words for Test-Case(s) Authoring

To make Test-Case(s) authoring more effective and simple “Drop-Down” list(s) are created as keywords for “Instruction” and “Entity Type”. The below tables contains all the available keywords and its functionality:

Test Case Editor	Item	Function
Action	Check	To verify the result in expected results
	Set	To set value(s) to parameters

	Wait	To keep waiting for ms or sec at a particular test-step
	While	To use for loops to repeat test steps for a particular count
	Run	To run a test case
	Repeat	To repeat multiple test-step(s) To repeat

Table 7: Drop-Down list of Instruction

Test Case Editor	Item	Function
Entity Type	CAN Signal	Available CAN signals from DBC file
	Simulation Input	To set signal in NVT App
	ECU Input	Available measurement from A2L file
	Simulation Output	To get signal from NVT App
	ECU Output	Available measurement from A2L file
	Time	To set/give time at a test step
	Test steps	Authored test step
	Ethernet Signal	Available Ethernet signals from arxml file
	LIN Signal	Available LIN signals from LDF file
	Test Case	Authored Test Cases

Table 8: Drop down list of Parameter

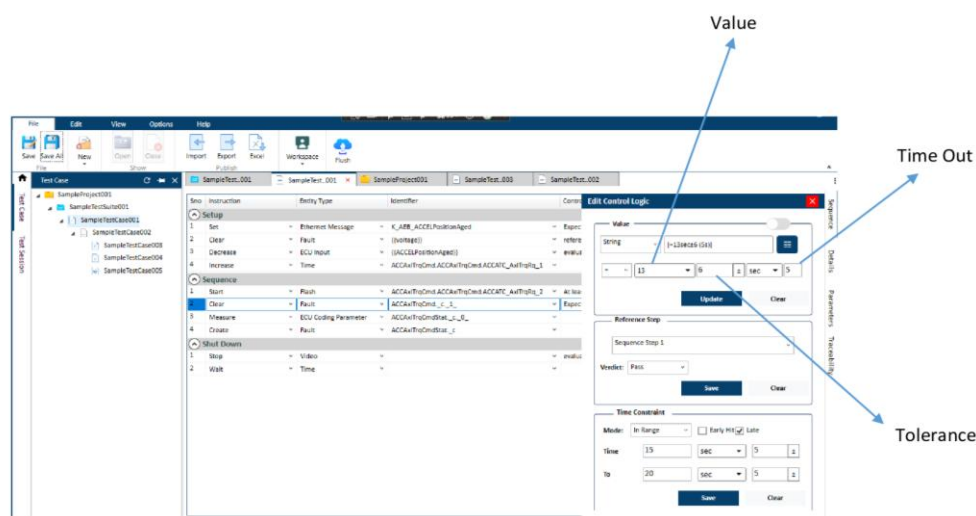
14.6 Identifier

- The Identifier field is a text input with autocomplete support that allows users to select or define the specific object targeted in a test step.
- The suggestion list is dynamically filtered based on the selected **Entity Type**.
- Users can type to search existing identifiers or enter a new identifier if it does not already exist (for example, a signal name, variable, or parameter).
- This field is used to precisely identify the message, signal, variable, or parameter that the test step will act upon.

- On dropdown selection or pressing **Enter** in the **Identifier** column, the system shall display a list of all signal names parsed from Signia database files (DBC, A2L, LDF) associated with the active Test Suite, Parameters, already used values, user defined data.

14.7 Control Logic

The **Control Logic** window allows users to define **conditions, references, and timing constraints** that control how an automated test step is executed and evaluated. This logic determines whether a step should pass, fail, or wait based on defined criteria. The **clear** option clears **only the updated or modified values** of the existing control logic and restores the previously saved configuration.



Snp.16 – Edit Control Logic Window

14.7.1 Value

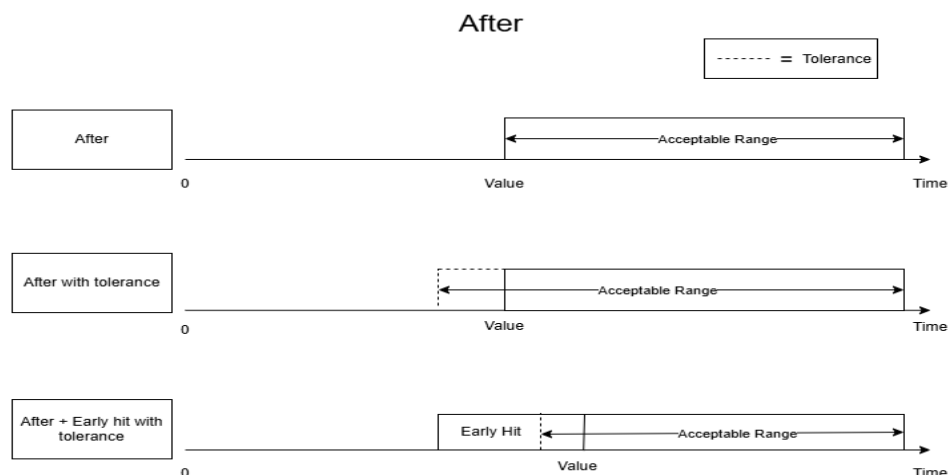
- The **Value** section is used to define the expected value or condition that must be satisfied during automated execution.
- Users can select the **data type** (for example, String or numeric type) and enter the expected value.
- A **comparison operator** can be selected to evaluate the actual runtime value against the expected value.
- Tolerance** and **unit** fields allow flexibility in value comparison, supporting range-based validation.
- This condition is evaluated automatically during test execution.

14.7.2 Reference Step

- The **Reference Step** section allows the current step to be evaluated based on the result of another step in the test sequence.
- Users can select a previously defined step as a reference.
- The **Verdict** field specifies the expected result of the referenced step (for example, Pass).
- During automation, the current step is executed or validated only if the referenced step meets the defined verdict.

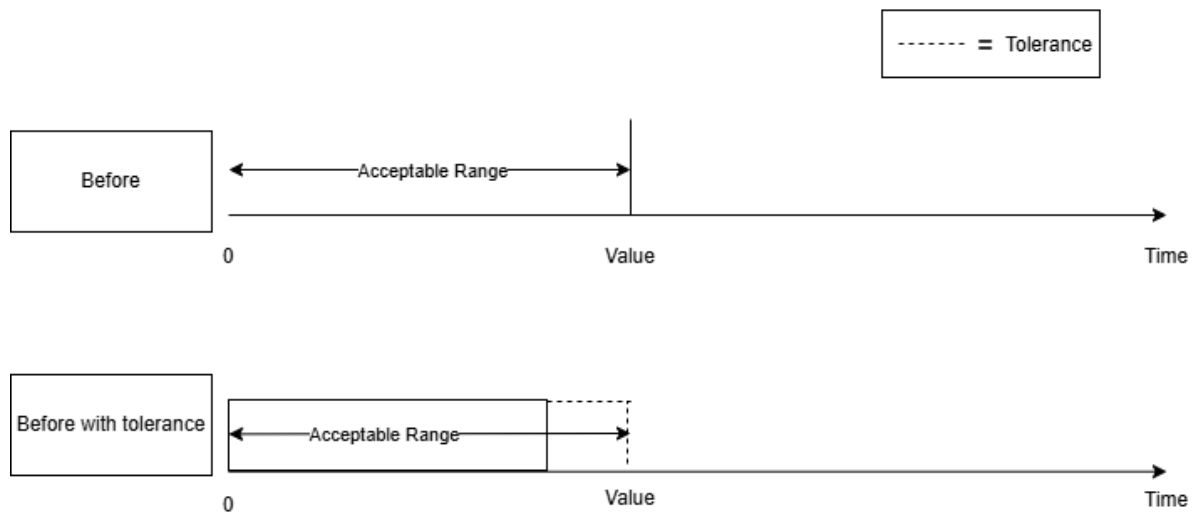
14.7.3 Time Constraint

- The **Time Constraint** section defines timing rules for automated execution.
- The **Mode** field specifies how timing should be applied (for example, timeout or delay).
- The **Time** field defines the duration for which the automation should wait or monitor the condition
- **Tolerance** allows minor timing variations without causing test failure.
- This ensures synchronization and reliable execution in time-dependent scenarios.
- There are 4 modes in time that was described below :
- **After**: The condition is considered valid only when it occurs after the acceptable range. With tolerance applied, the condition may be accepted even if it occurs slightly before the end of the acceptable range, or within a defined early tolerance window.

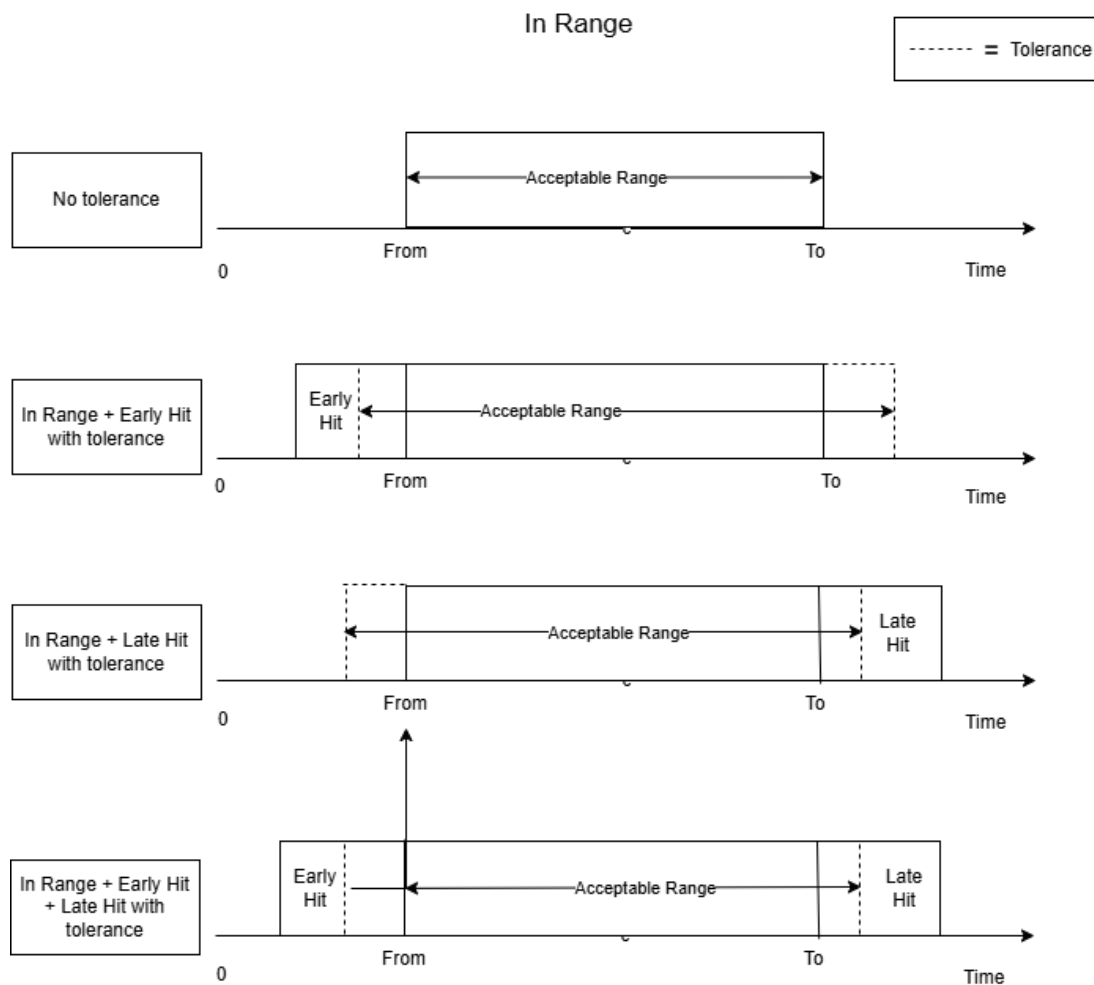


- **Before**: The condition is considered valid only when it occurs before the acceptable range. When tolerance is enabled, the condition may still be treated as valid if it slightly overlaps into the beginning of the acceptable range, within the defined tolerance limits.

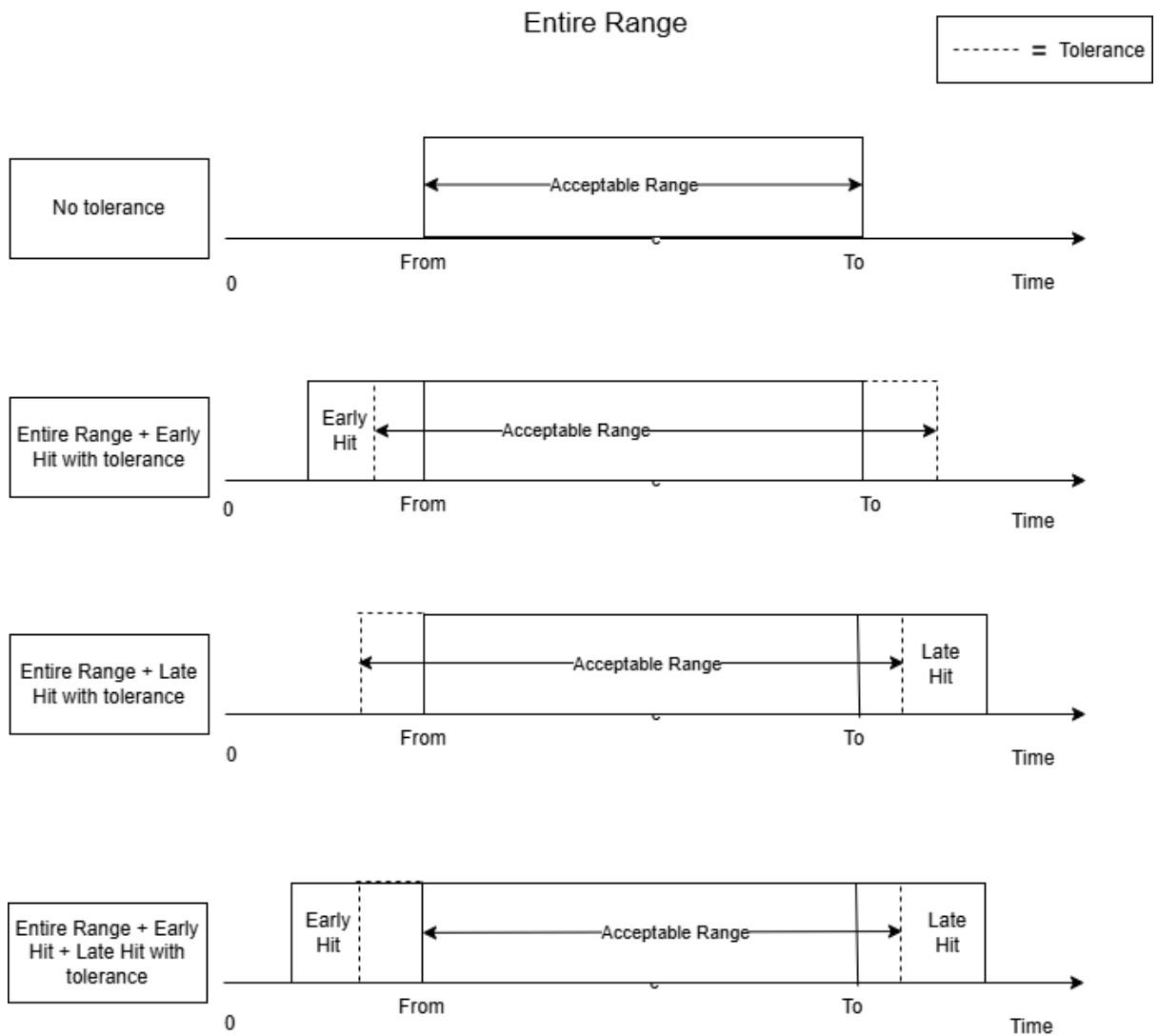
Before



- **In Range:** The condition is considered valid when it occurs within the acceptable range. Depending on the configuration, limited tolerance may allow the condition to be satisfied slightly before the start of the range (early hit), slightly after the end of the range (late hit), or on both sides. If no tolerance is applied, the condition must strictly stay within the defined range.



- **Entire Range:** The condition is same as In Range but the difference is that the value should occur in the entire time range. The condition is considered valid when it occurs anywhere within the acceptable range. When tolerance is enabled, the condition may also be accepted if it occurs slightly before or slightly after the acceptable range, as long as it remains within the configured tolerance boundaries.



- Example of Time Constraint: In the below image the condition is evaluated over the entire time range from **5 to 10 seconds**. With **Early Hit** and **Late** enabled and a tolerance of **±1 second**, the condition is accepted if it occurs **between 4 and 11 seconds**.

Time Constraint

Mode: Entire Range ☒ Early Hit ☒ Late

Time sec ±

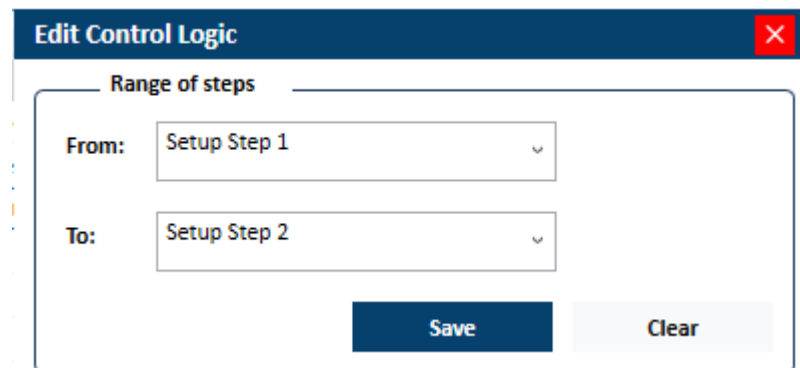
To sec ±

Save Clear

Snv.17 – Time Constraint Example

14.7.4 Range of Steps

- This control logic is available only when the “When” action is selected in the instruction and is used to evaluate conditions over a specified range of previously authored steps.
- For example, if the range is set from step 1 to step 2, the conditions defined in the While logic are continuously checked during the execution of steps 1 through 2.
- In the tool, this range is displayed as “From Setup Step 1 to Setup Step 2”, indicating that the condition is evaluated during the execution of those setup steps.



The screenshot shows a dialog box titled "Edit Control Logic" with a red close button in the top right corner. Inside the dialog, there is a section titled "Range of steps". Under this section, there are two dropdown menus. The first is labeled "From:" and is set to "Setup Step 1". The second is labeled "To:" and is set to "Setup Step 2". At the bottom of the dialog, there are two buttons: "Save" and "Clear".

Snp.18 – Range of Steps for While

- This range of steps is also available when the Repeat action is selected.
- When Repeat is enabled, additional automation logic is provided to control how the steps are repeated:
 - Condition defines the criteria that must be satisfied for the repeat execution to continue.
 - Count specifies the maximum number of times the selected range of steps should be executed.
 - Duration specifies the maximum time for which the selected range of steps should be repeated, after which the repeat execution stops.
- The selected setup steps **from Setup Step 1 to Setup Step 3** are executed repeatedly. The repetition continues **until the defined condition is met** and is controlled by **duration**, meaning the steps will keep repeating for **5 ms**, after which the execution stops automatically.

Edit Control Logic

Range of steps

From: Setup Step 1

To: Setup Step 3

Condition :
Until

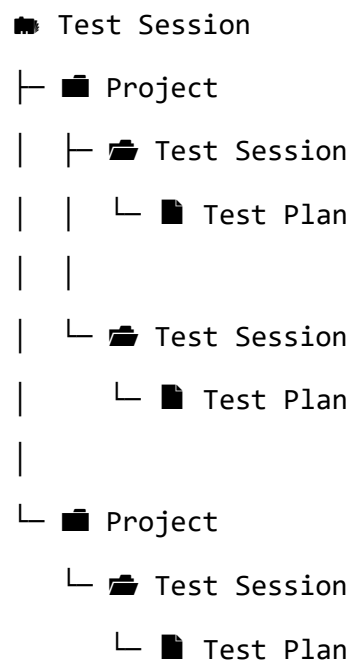
☐ Count
☒ Duration

Duration Value :
5
ms

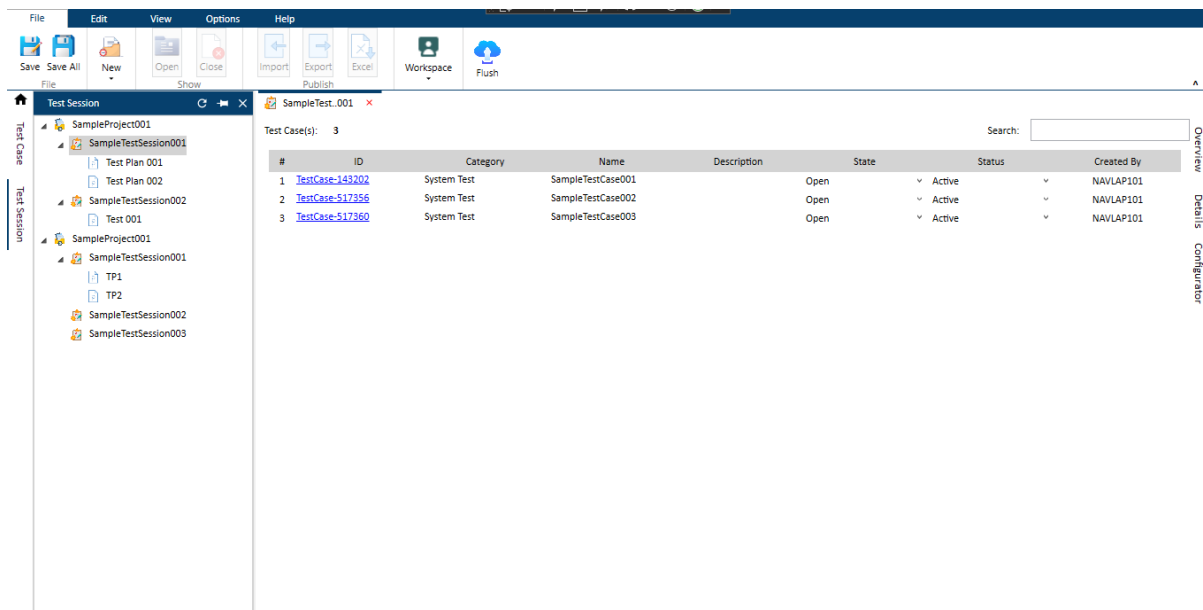
Save
Clear

Snp.19 – Range of Steps for Repeat

15 Test Session Hierarchy



Snp.20 - Hierarchical Structure of Test Session



SnP.21 – Test Session Hierarchy

16 Test Case Execution Workflow

Once test cases are authored in the **Test Case** tab, they must go through the following execution workflow:

- Link the test cases with a Test Session
- Test Bench Configuration setup
- Define and configure a Test Plan
- Execute the configured test cases

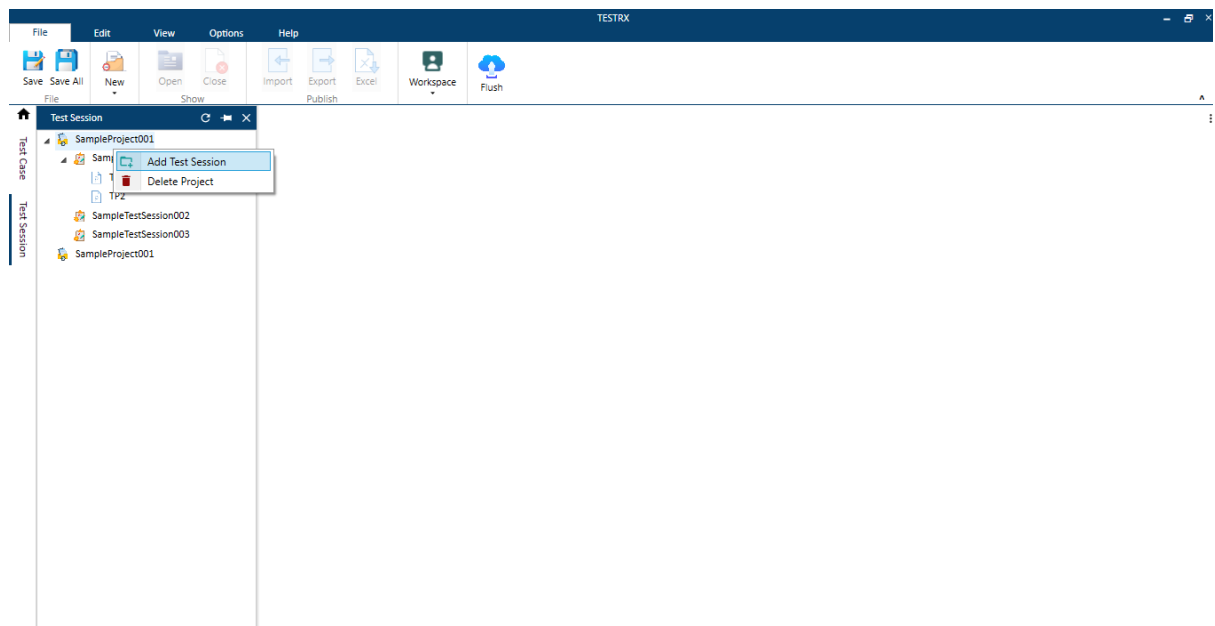
This structured process ensures controlled planning, traceability, and systematic execution of test cases within the application.

16.1 Link Test Cases with a Test Session

This section describes the procedure for associating test cases with a test session to enable execution and tracking. To view the test steps in Test Session or Test Plan click on the ID column of that particular test case.

16.1.1.1 Select a Project and Create a Test Session

- From the **File** menu, select **New** and hover over **Select Project**. The system displays all projects available in the **Test Case** tab. Select the required project to add it to the workspace.
- After the project is added, select the project node right click and choose **Add Test Session**. Alternatively, a test session can be created by selecting **Test Session** from the **New** option in the **File** menu.

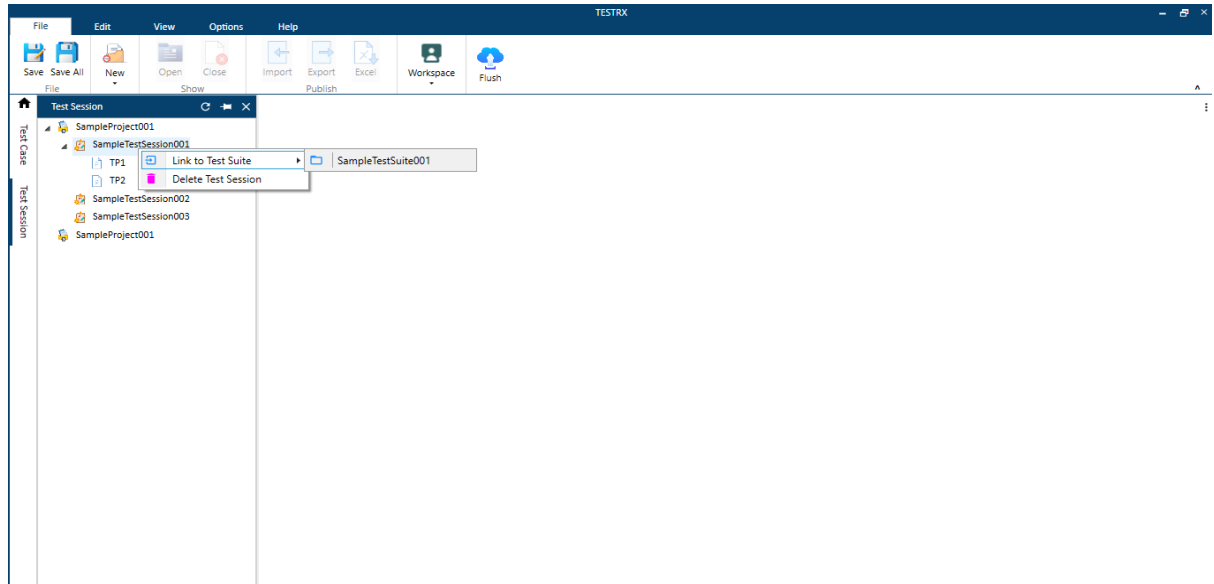


Snp.22 – Add Test Session

16.1.2 Link Test Suites and Add Test Cases

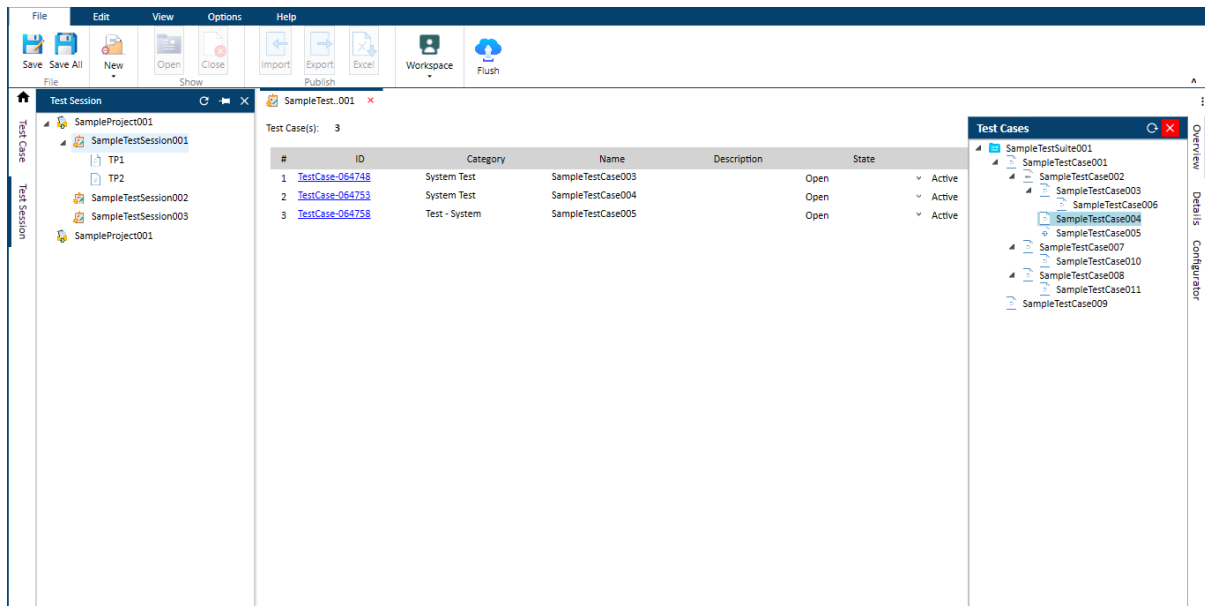
- Once the test session is created, right-click the test session and select **Link to Test Suite**.

The system displays a list of all available test suites. Select the appropriate test suite.



Snp.23 – Link to Test Suite

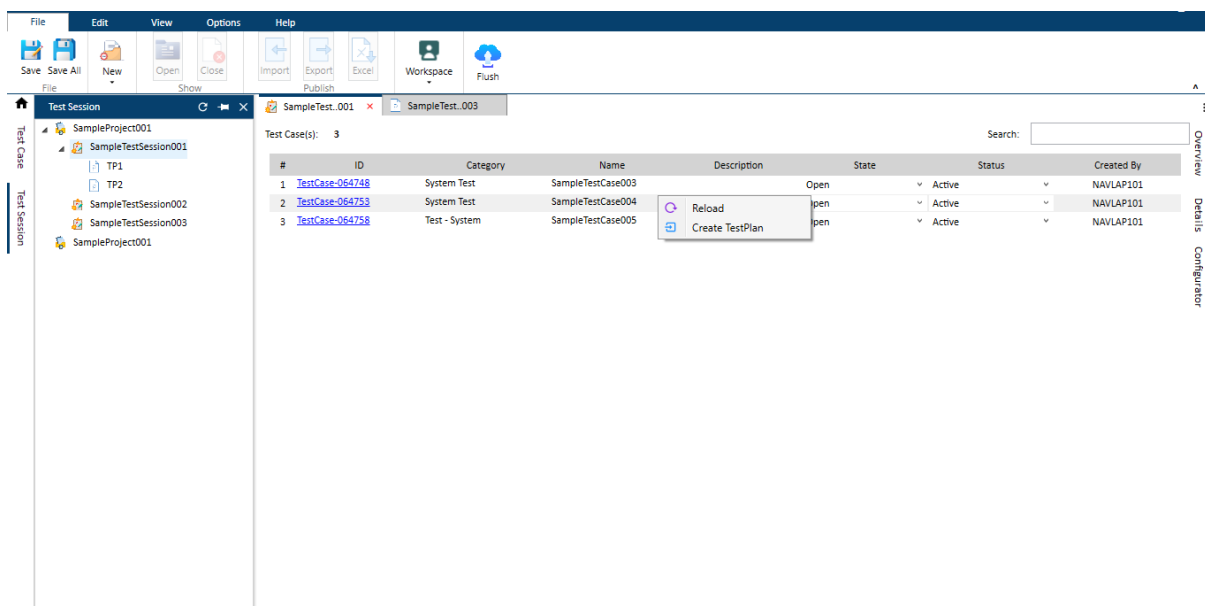
- After a test suite is selected, the **Test Cases** panel is displayed on the right side, listing all test cases associated with the selected test suite.
- Expand the test suite using the arrow icon to view individual test cases. Drag and drop the required test cases into the test session to include them for execution.
- When a nested test case is collapsed and dragged into a test session, all its child test cases are automatically included in the test session.
- To include test cases from additional test suites, repeat the same procedure by linking the required test suite to the test session.



Snp.24 – Window for drag and drop of test cases

16.1.3 Synchronize and Manage Test Cases

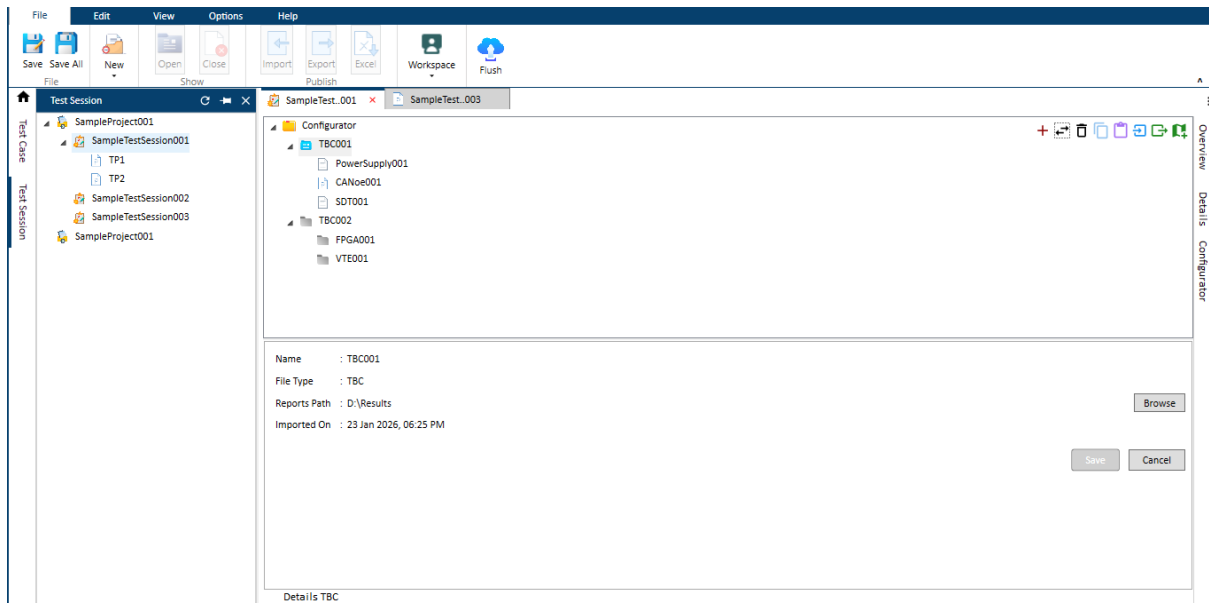
- If changes are made to test cases in the **Test Case** tab, use the **Reload** option to synchronize the updates. This option is available both in the **Test Cases** panel and through the right-click menu within the **Test Session** table.
- Within the **Test Session** tab, users can update the state and execution status of the linked test cases as required, enabling accurate tracking of execution progress and results.



Snp.25 – Reload Option

16.2 Test Bench Configuration Setup

After linking test cases to a test session, users must configure **Test Bench Configuration (TBC)** details to enable execution on a designated test bench. TESTRX requires these details to correctly initialize, control, and manage the hardware and simulation environments associated with the selected bench during project execution.



Snp.26 – Configurator Tab

16.2.1 Procedure to Add Configuration Details

1. Open the Configurator tab within the required Test Session.
2. Select the Configurator folder corresponding to the test session to be executed.
3. Click the Add (+) icon in the upper panel and select Add TBC.
4. Choose the appropriate path for the reports to be stored.
5. Click the Add (+) icon again and select the required interface from the available options.

Component	Description
CANoe	Used for network communication and signal simulation during automated testing.
PowerSupply	Controls power settings and power cycling for test execution.
SDT	Supports execution of service diagnostic tool.
FPGA	Enables interaction with FPGA-based hardware components.
VTE	Provides a virtual environment for automated test execution.

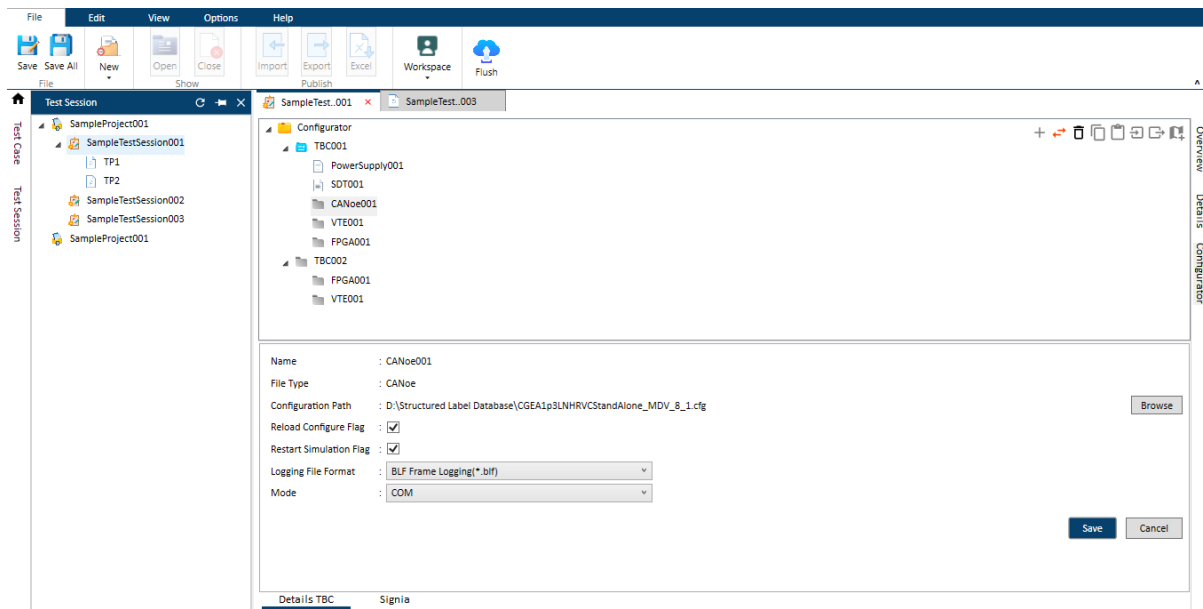
Table 9:Configuration Components

16.2.2 Supported Interfaces and Configuration Parameters

1. CANoe

Allows configuration of CANoe-related execution settings, including:

- Configuration file path
- Logging file format
- Reload configuration option (enable/disable)
- Restart simulation option (enable/disable)
- Mode selection: **XIL** or **COM**
- DBC files can be associated through the **Signia** tab

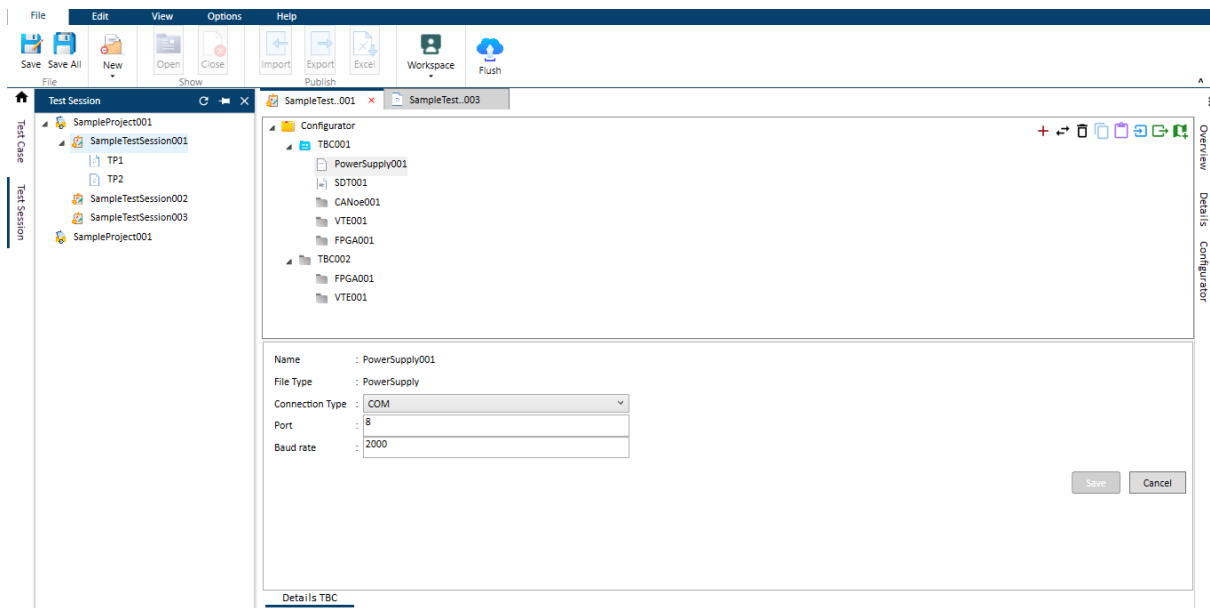


Snp.27 – CANoe Configuration

2. Power Supply

Enables setup of power control parameters:

- Connection type selection: **PyVISA** or **COM**

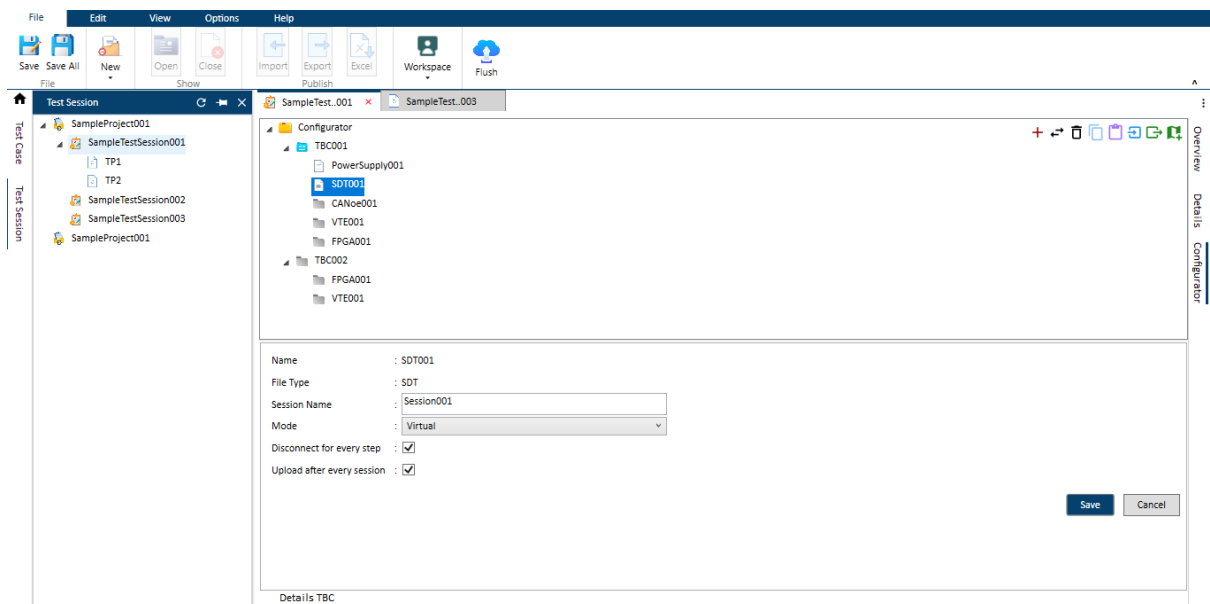


Snp.28 – Power Supply Configuration

3. SDT

Supports diagnostic and session-level configuration:

- Session name definition
- Mode selection
- Option to disconnect after each step (enable/disable)
- Option to upload data after each session (enable/disable)

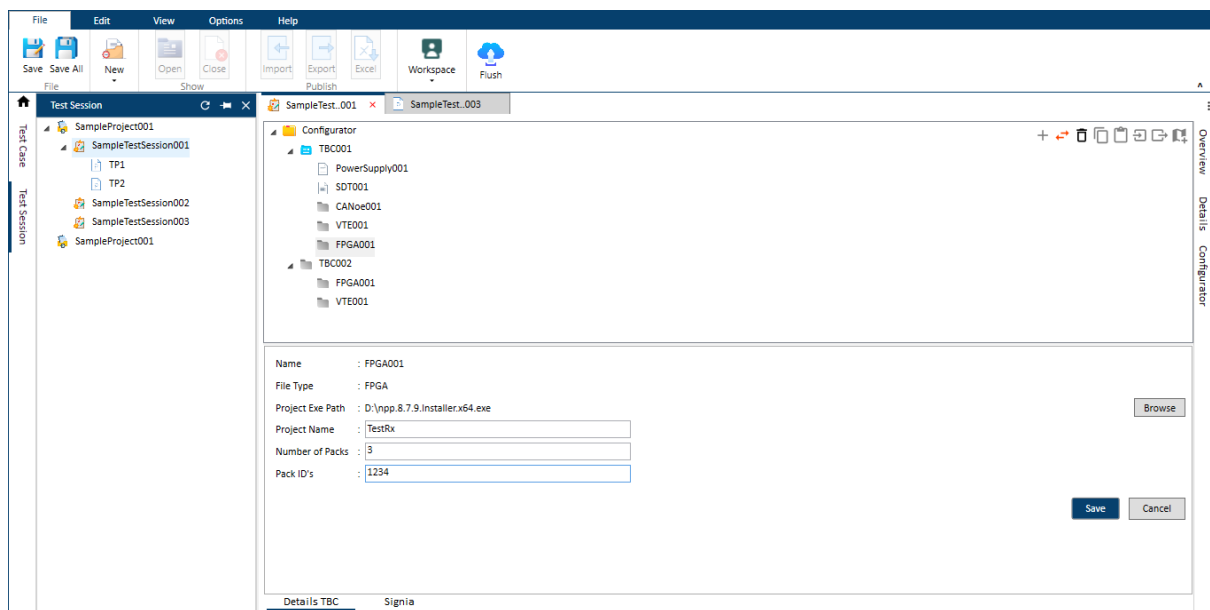


Snp.29 – SDT Configuration

4. FPGA

Used for FPGA-based execution configuration:

- Executable (.exe) file selection
- Project name
- Number of packs
- Pack IDs
- CSV files can be associated through the **Signia** tab

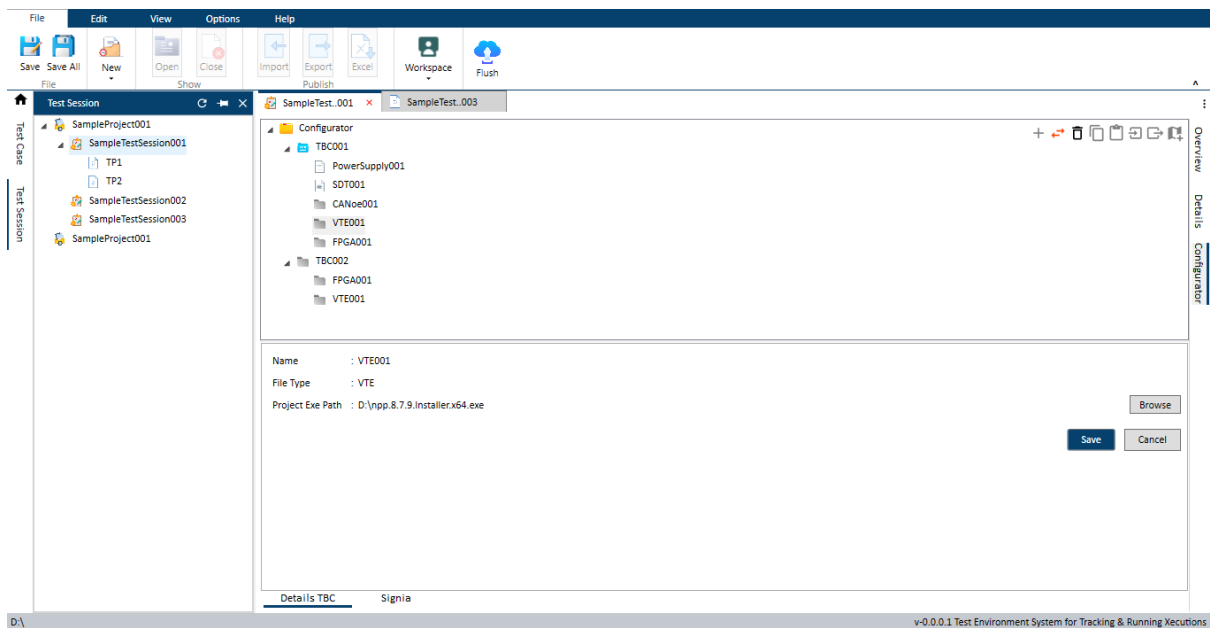


Snp.30 – FPGA Configuration

5. VTE (Virtual Test Environment)

Used for configuring and managing the Virtual Test Environment required for automated execution:

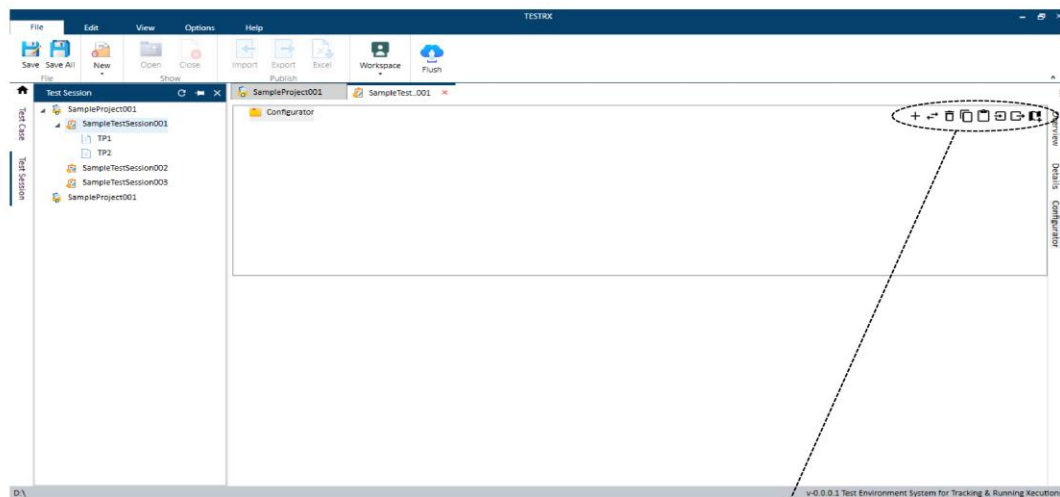
- Executable (.exe) file selection
- VTE project name
- File path configuration
- Import and management of VTE database files
- VTE files can be associated and managed through the **Signia** tab



Snp.31 – VTE Configuration

16.2.3 Configurator Tab – Toolbar Options

The Configurator tab provides a set of toolbar actions that allow users to efficiently manage **Test Bench Configuration (TBC)** items. These options are available in the upper panel of the Configurator tab and support standard configuration management activities.



Icon	Description
	To add TBC and supported interfaces
	Enable a TBC
	Delete a TBC
	Copy a TBC
	Paste a TBC
	Import a TBC
	Export a TBC
	Map Labels

Snp.32 – Configurator Tool bar

The following actions are available:

- **Add:** Adds a new item, such as a Test Bench Configuration (TBC) or an interface, under the selected node.
- **Enable TBC:** Enables the selected TBC node for execution.
- **Delete:** Deletes the selected TBC node from the configurator.
- **Copy:** Creates a duplicate of the selected configuration item for reuse.
- **Paste:** Pastes the copied configuration item into the selected location within the configurator hierarchy.
- **Import:** Imports a previously exported TBC ZIP file into the configurator.
- **Export:** Exports the selected TBC as a ZIP file for backup or reuse.
- **Map Labels:** Allows users to define and manage read and write paths for Test and I/O labels in the Map Labels Window.

16.2.4 Map Labels

- The **Map Labels** option is available only for **TBC nodes** in the project tree.
- Selecting this option opens the Map Labels window, allowing users to view, edit, import, export, and save label mappings.
- The **Map Labels Window** provides a centralized interface for configuring and managing label-to-signal mappings used during test execution.
- It allows users to map logical labels defined in test cases to their corresponding database signals, ensuring correct read/write and diagnostic communication during execution.

Map Labels Window

Test Labels
IO Labels

Sno	Name	Read Path	Write Path
1	K_AEB_ACCELPositionAged	K_AEB_ACCELPositionAged	K_AEB_ACCELPositionAged
2	ACCxITrqCmd.ACCxITrqCmd.ACC	ACCxITrqCmd.ACCxITrqCmd.ACCATC_AxITrqRq_1	ACCxITrqCmd.ACCxITrqCmd.ACCATC_AxITrqRq_1
3	ACCxITrqCmd.ACCxITrqCmd.ACC	ACCxITrqCmd.ACCxITrqCmd.ACCATC_AxITrqRq_2	ACCxITrqCmd.ACCxITrqCmd.ACCATC_AxITrqRq_2
4	ACCxITrqCmd_c_1_	ACCxITrqCmd_c_1_	ACCxITrqCmd_c_1_
5	ACCxITrqCmdStat_c_0_	ACCxITrqCmdStat_c_0_	ACCxITrqCmdStat_c_0_

Label : K_AEB_ACCELPositionAged
☐ Dependencies

Read Path	K_AEB_ACCELPositionAged	Write Path	K_AEB_ACCELPositionAged
Read Path Tool Interface	CANoe	Write Path Tool Interface	CANoe
Read Path File Name	A2L001	Write Path File Name	A2L001

Details

Autofill
Cancel
Save

Snp.33 – Map Labels Window

16.2.4.1 Overview

- The Map Labels window consists of two main tabs:

Test Labels

IO Labels

- The Test Labels tab is displayed by default when the window is opened.
- Common action buttons — **Autofill**, **Import**, **Export**, **Cancel**, and **Save** — are available at the bottom of the window.
- A **details panel** is displayed below the grid to show additional read/write path information for the selected label.
- The **Dependencies** option allows users to add dependency labels when required. If no dependencies are needed, this option can be disabled.

16.2.4.2 Test Labels Tab

- The **Test Labels** tab displays all labels used in the **current test session**. These labels are automatically extracted from the test cases linked to the active test session.

- This tab enables mapping of test labels to their corresponding **Read Path** and **Write Path** sourced from the active **CANoe database**.
- The tab is divided into two vertically stacked sections:
 - Test Label Mapping Grid (Upper Section): The grid automatically lists all test labels used in the active test session. Labels are dynamically collected by analysing all test cases and extracting label definitions from their Labels objects.
 - Detailed Mapping Information Panel (Lower Section): This section displays detailed, **read-only** information for the selected label, including Full Read Path and Write Path, Tool interface used (e.g., CANoe), Source database file names. The details panel automatically updates when a different label is selected in the grid.

16.2.4.3 IO Labels Tab

- The **IO Labels** tab provides a unified interface for mapping all **user-defined input/output labels** to database Read and Write paths. These labels represent hardware or external interfaces used during test execution.
- The tab consists of two sections:
 - I/O Mapping Grid (Upper Section): Displays all IO labels extracted from the test session, with editable Read and Write path mappings. Invalid or missing signal mappings are highlighted visually to indicate configuration issues.
 - Mapping Details Panel (Lower Section): Displays tool interface details, file references, and mapping information for the selected IO label. All fields in this section are **non-editable** and update automatically based on grid selection.

16.2.4.4 Autofill Functionality

The **Autofill** option automatically populates Read and Write paths across all three tabs — **Test Labels**, **IO Labels**, and **Diagnostics** — by matching label names with available signals from connected CANoe databases.

16.2.4.5 Import and Export

- The **Import** option is used to load label mapping data for all tabs from a ZIP file. If mapping data already exists, a confirmation is displayed before replacing the current mappings.

- The **Export** option is used to save the current label mappings from all tabs into a ZIP file. The export operation does not affect the existing mappings in the workspace.

16.2.4.6 Label Validation and Visual Indicators

- Labels with missing or invalid database signals are highlighted in pink.
- Read/Write dropdowns are disabled when valid signals are unavailable.
- All dropdowns display full hierarchical signal names and refresh automatically when the database or configuration changes.

16.3 Define and Configure a Test Plan

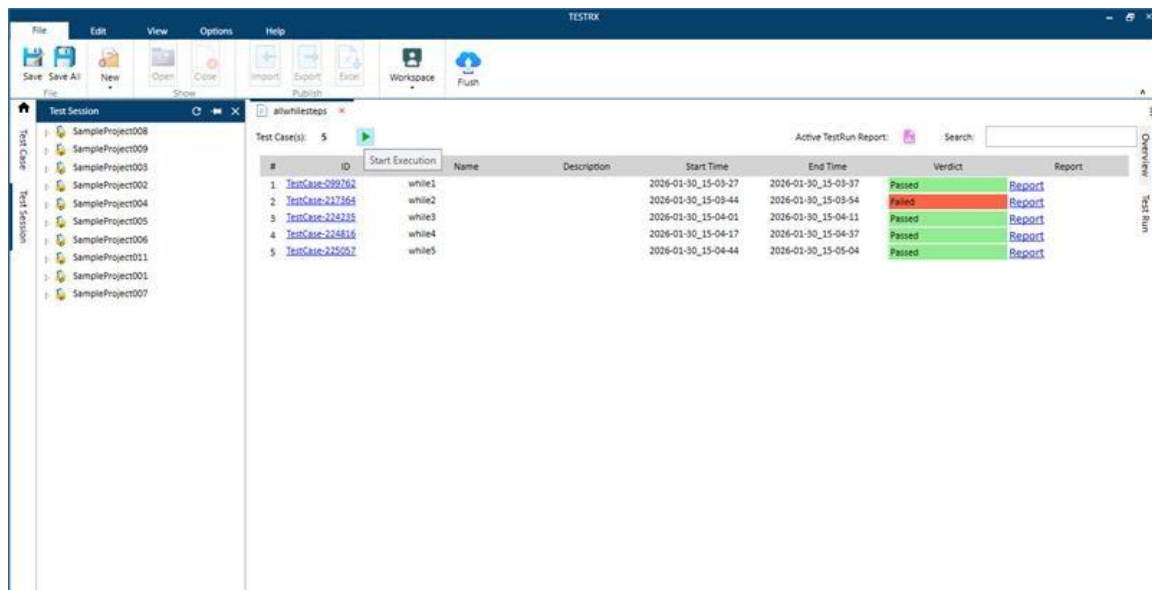
- After adding the Test Bench Configuration (TBC), return to the Test Session Overview tab.
 1. Select the test cases to be executed.
 2. Right-click on the selected test cases and choose Create Test Plan.
 3. Enter the required Test Plan name.
- A new test plan is then created under the selected test session, using the name provided in the previous step.
- Additionally, a Reload option is available in the Test Session Overview tab to synchronize and reflect the latest updates from the Test Case tab.

Note: Use **Ctrl + A** to select all test cases, then right-click to create a test plan or reload the test cases.

16.4 Execute the configured test cases

- When a created **Test Plan** is double-clicked, a dedicated tab is opened displaying the execution details of the test plan.
- This tab presents a structured, tabular view of the test cases with the following information:
 - **S. No** – Sequential number of the test case within the session
 - **ID** – Unique identifier assigned to the test case
 - **Name** – Name of the test case
 - **Description** – Brief description of the test case
 - **Start Time** – Timestamp indicating when execution started
 - **End Time** – Timestamp indicating when execution completed
 - **Verdict** – Execution result status of the test case
 - **Report** – Link to the generated execution report, if available
- The **Test Case(s)** counter at the top of the tab indicates the total number of test cases included in the selected test plan.

- The **Start Execution**(▶) button allows users to initiate execution of the entire test plan or the selected test cases in test plan.

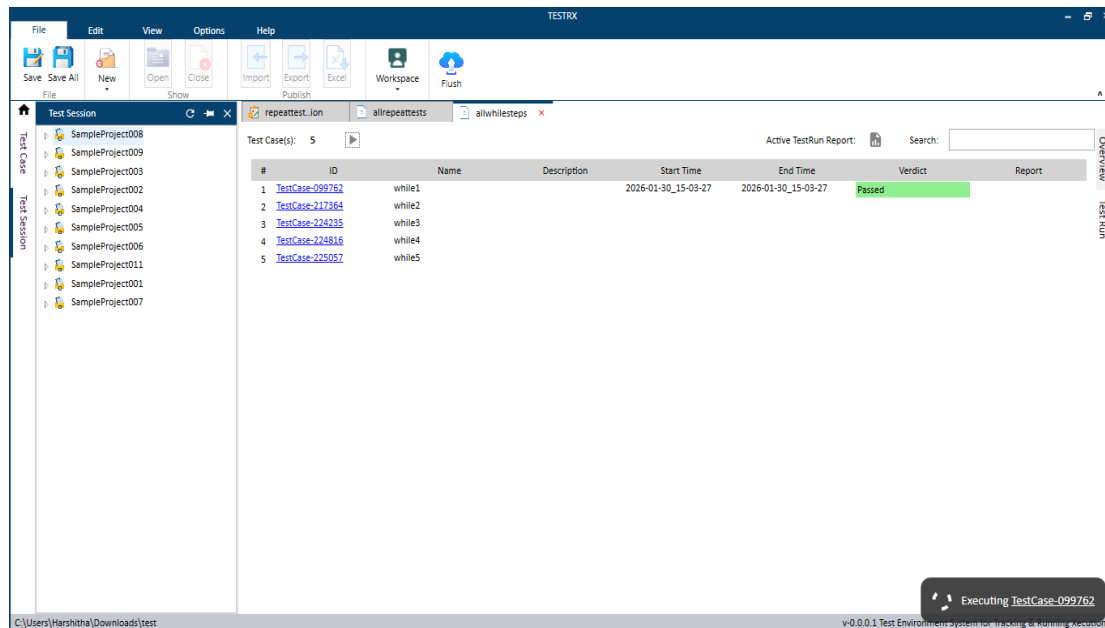


Snip.34 – Start Execution Icon

- An **Active Test Run Report** indicator provides quick access to the most recent execution html report, while the **Search** field enables efficient filtering of test cases within the tab.
- In the **Test Run** tab of the test plan, users can view the results of the most recent test execution.

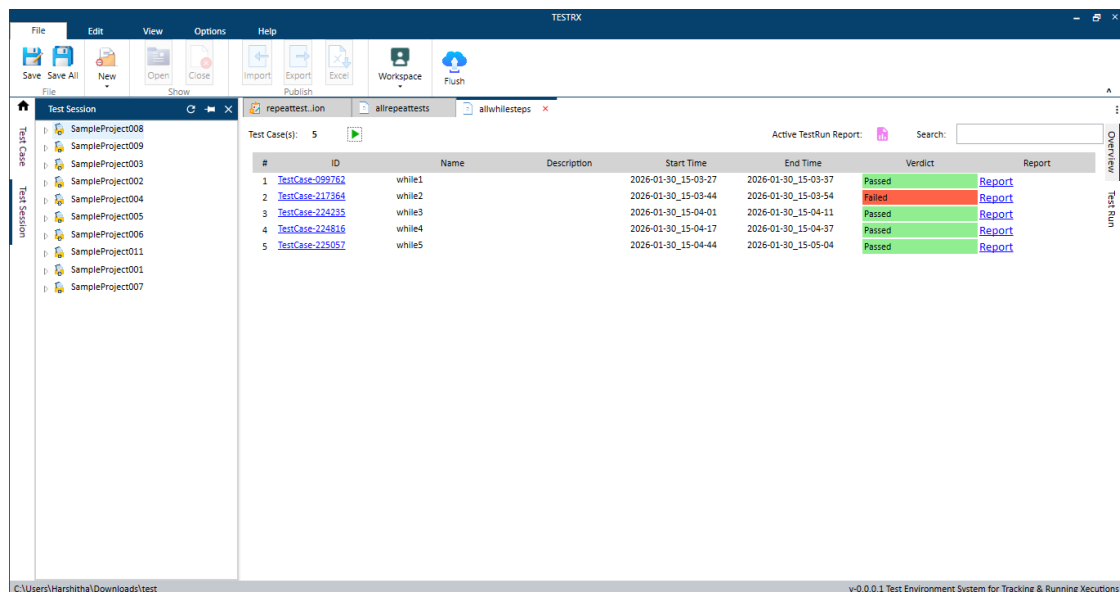
16.5 Test Plan Execution and Report Viewing

- When Start Execution is clicked, the execution of the test plan begins, as shown in the image below.



Snip.35 – Test Plan Execution

- After execution of all test cases is completed, the result of each test case is displayed in the Verdict column.



Snip.36 – Verdict of each test case

- When the ID column of a particular test case is clicked, the execution results of each test step for that test case are displayed.

The screenshot shows the TESTRX application window. On the left is a sidebar with a 'Test Session' tree containing various sample projects. The main area displays a table for the 'while1' test session. The table has columns for Sno, Instruction, Entity Type, Identifier, Control Logic, Actual Value, and Verdict. The data shows a sequence of 7 steps, all of which passed.

Sno	Instruction	Entity Type	Identifier	Control Logic	Actual Value	Verdict
1	Set	CAN Signal	GearBoxInfo Gear	Expected Value is +1-Gear_1 (time: 1		Passed
2	Check	CAN Signal	GearBoxInfo Gear	Expected Value is +1-Gear_1 (time: 1		Passed
3	Set	CAN Signal	GearBoxInfo Gear	Expected Value is +2-Gear_2 (time: 2		Passed
4	Check	CAN Signal	GearBoxInfo Gear	Expected Value is +2-Gear_2 (time: 2		Passed
5	Wait	Time		Expected Value is +10 (timeout:10	10	Passed
6	While	Test Step(s)		From Sequence Step 2 To Sequence		Passed
6.1	Set	CAN Signal	GearBoxInfo Gear	Expected Value is +2-Gear_2 (time: 2		Passed
6.2	Check	CAN Signal	GearBoxInfo Gear	Expected Value is +2-Gear_2 (time: 2		Passed
7	Check	CAN Signal	GearBoxInfo Gear	Expected Value is +2-Gear_2 (time: 2		Passed

Snp.37 – Verdict of each test step

- The respective test report for each test case is available in the Report column. Clicking the report link opens the latest report for that test case in the browser.

The screenshot shows the 'TESTRX Test Report' for 'TestCase-099762 :: while1'. It is divided into three main sections: Global Information, Execution Information, and Execution Overview.

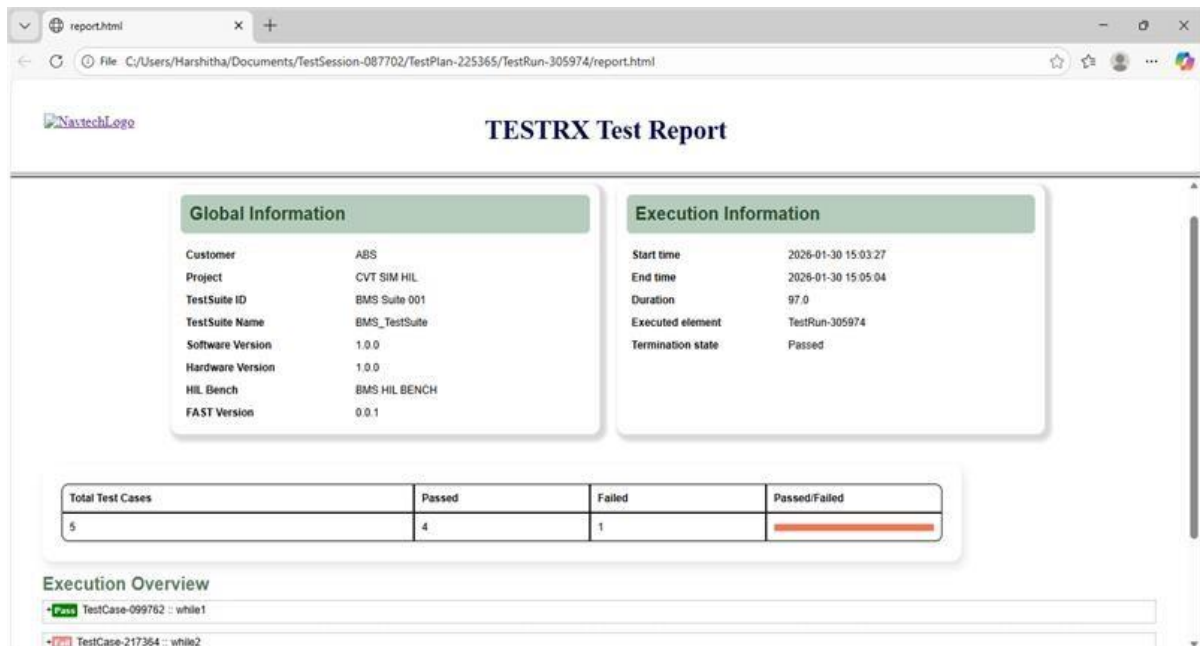
Global Information		Execution Information	
Customer	ABS	Start time	2026-01-30 15:03:27
Project	CVT SIM HIL	End time	2026-01-30 15:03:37
TestSuite ID	BMS Suite 001	Duration	10.0
TestSuite Name	BMS_TestSuite	Executed element	TestCase-099762
Software Version	1.0.0	Termination state	Passed
Hardware Version	1.0.0		
HIL Bench	BMS HIL BENCH		
FAST Version	0.0.1		

Execution Overview

+ Pass TestCase-099762 :: while1

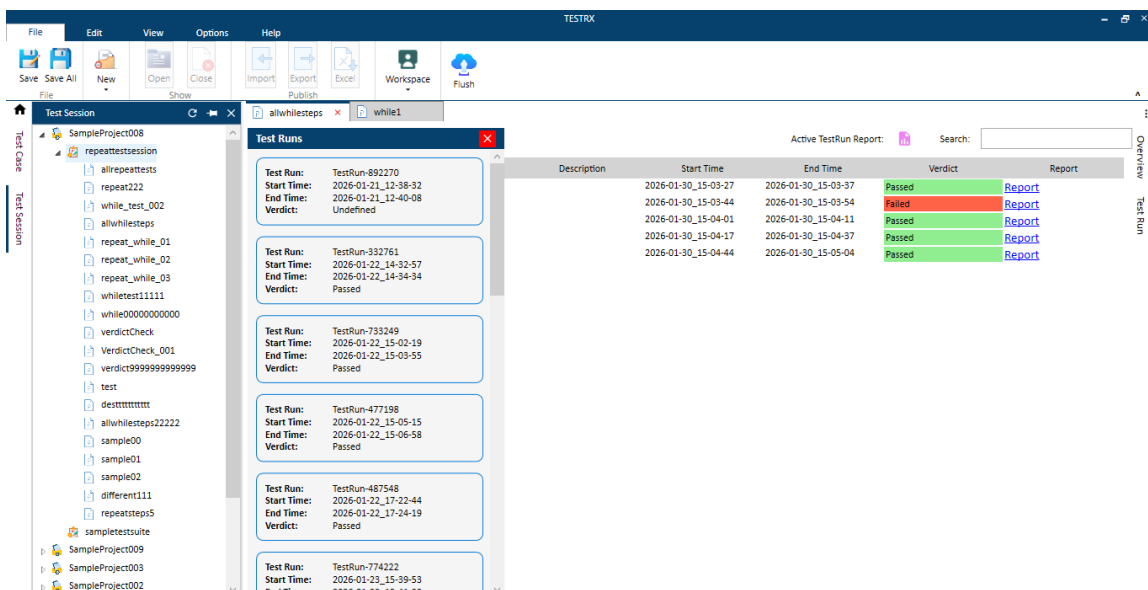
Snp.38 – Report of Individual Test Case

- When the report next to the Active Test Run report is clicked, the report for the entire latest test run is opened.



Snp.39 – Active Test Run Report

- To view older reports of the test plan, go to the Test Run tab and click on My Runs(≡). This displays a list of all previous runs. Clicking on a report opens the corresponding results.



Snp.40 – Previous run reports

17 JAMA Integration

Jama is an **Application Lifecycle Management (ALM)** tool used to manage requirements, test cases, and traceability.

The JAMA Integration enables **automation-based synchronization** between the test automation framework and Jama Connect, ensuring consistent and up-to-date test artifacts.

17.1 Test Case Role

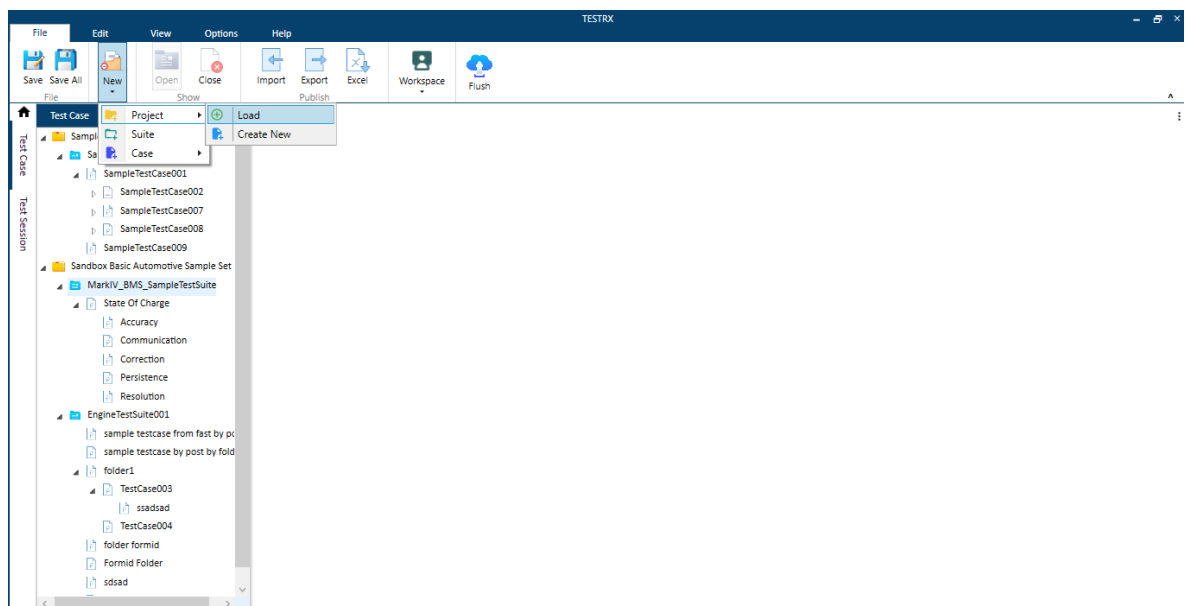
In the **Test Case role**, users can automate interactions with ALM to manage test projects, test suites, and test cases. This role supports loading projects from ALM, synchronizing automated test updates, and maintaining requirement traceability.

Key functionalities include:

- Loading of test projects from ALM
- Flushing modified test suites and test cases back to ALM
- Linking requirements to test cases through the Traceability tab

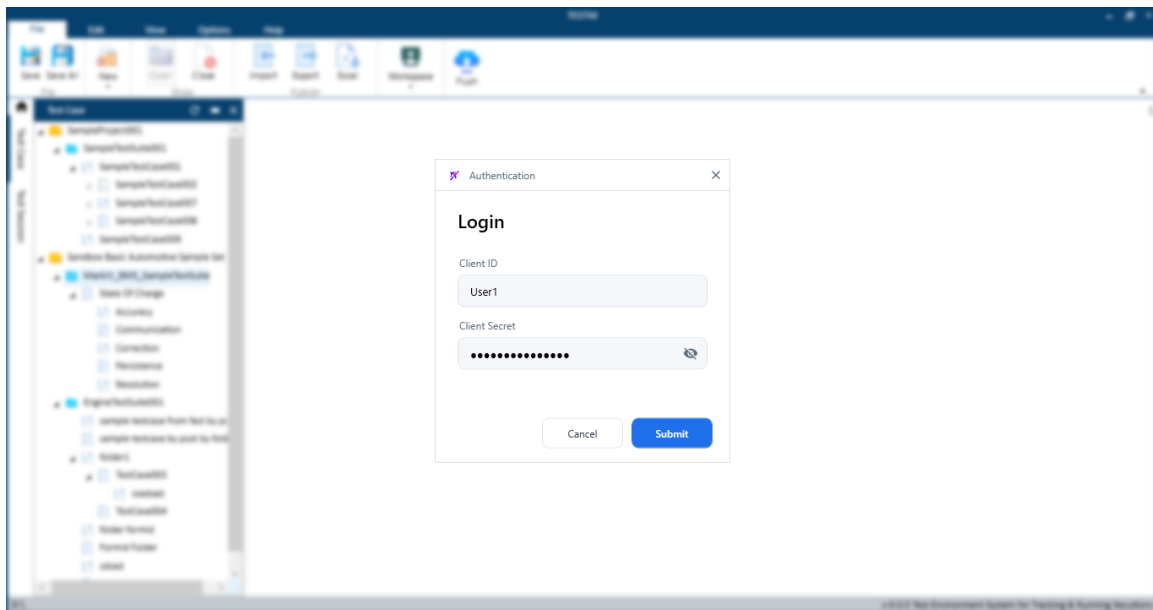
17.1.1 Load Project from ALM

- To load a project from ALM, navigate to File → New → Project and select the Load option.



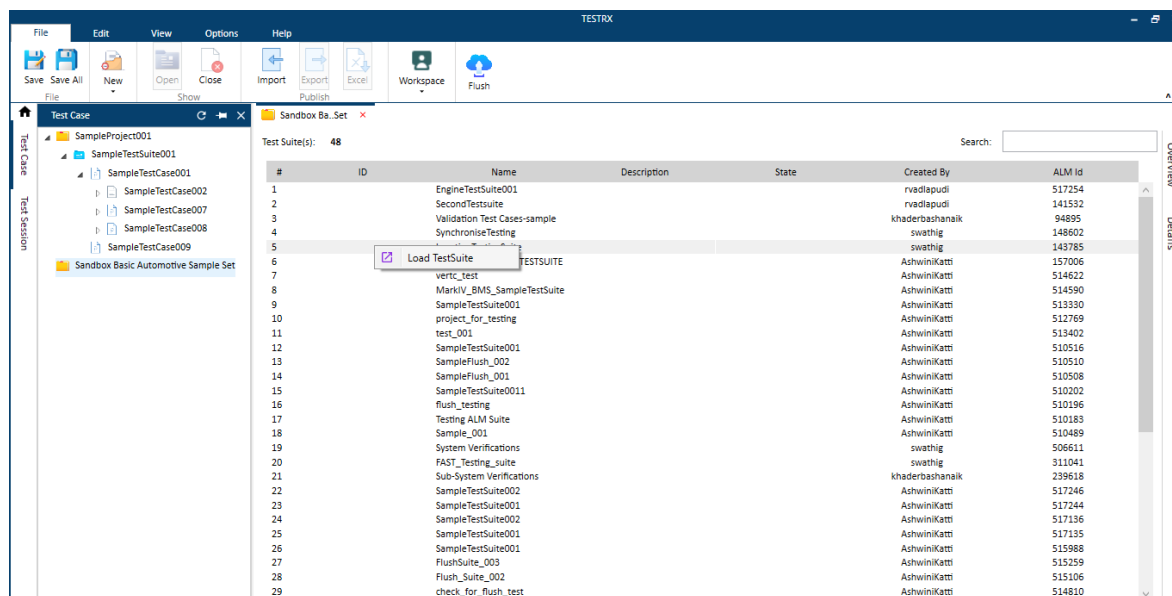
Snp.41 – Load option for project

- Enter valid ALM credentials and click Submit.



Snp.42 – Authentication Window

- The Load Project window is displayed. Select the required project from the drop-down list and click OK. The selected project is then loaded into the automation environment.
- After loading, double-click the project to view all available test suites associated with it. To load a specific test suite, select the required test suite from the list, right-click, and choose Load Test Suite.

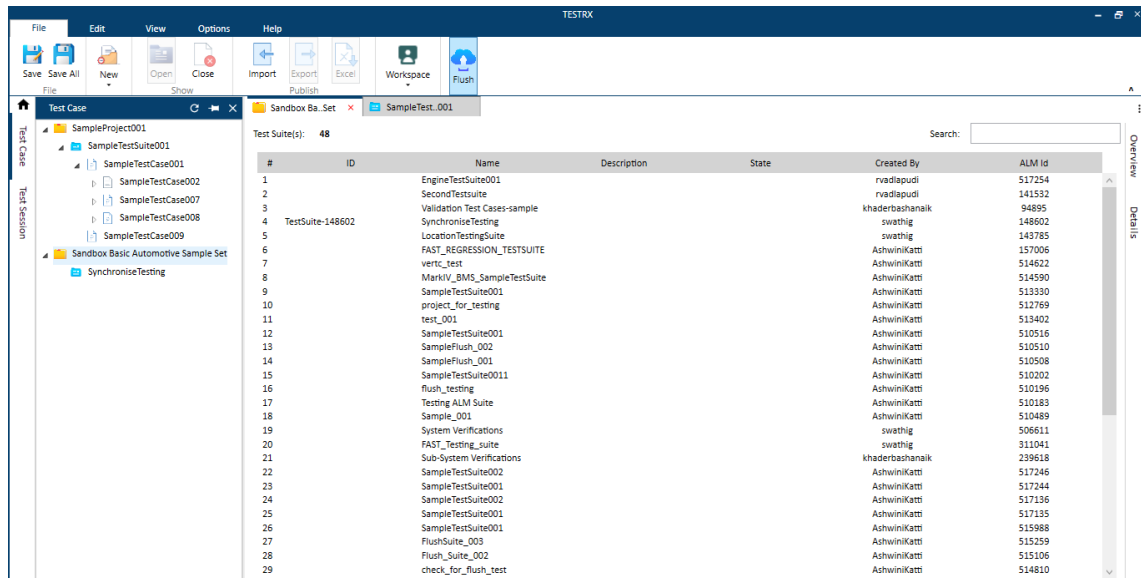


Snp.43 – Load Test Suite Option

- Provide valid credentials when prompted. The selected test suite is then loaded and becomes available for automation and execution.

17.1.2 Flush Modified Test Suites and Test Cases

- If any modifications are made to test suites or test cases that were loaded from ALM, those changes can be synchronized back to ALM using the Flush operation.
- To flush updates, select the modified test suite or test case and navigate to File → Flush. Enter valid ALM credentials and click Submit.



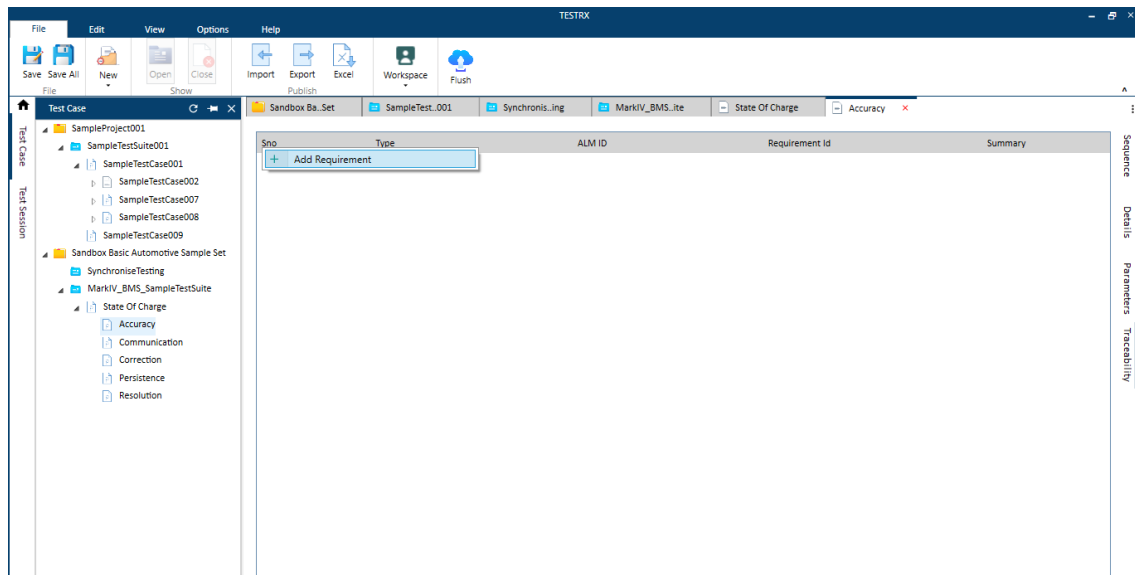
Snp.44 – Flush Menu Option

- The flushing progress and status are displayed in a notification tooltip at the top-right corner of the application. Options to minimize or close the notification are also available.

Note: Only test suites and test cases that were originally loaded from ALM are eligible for flushing.

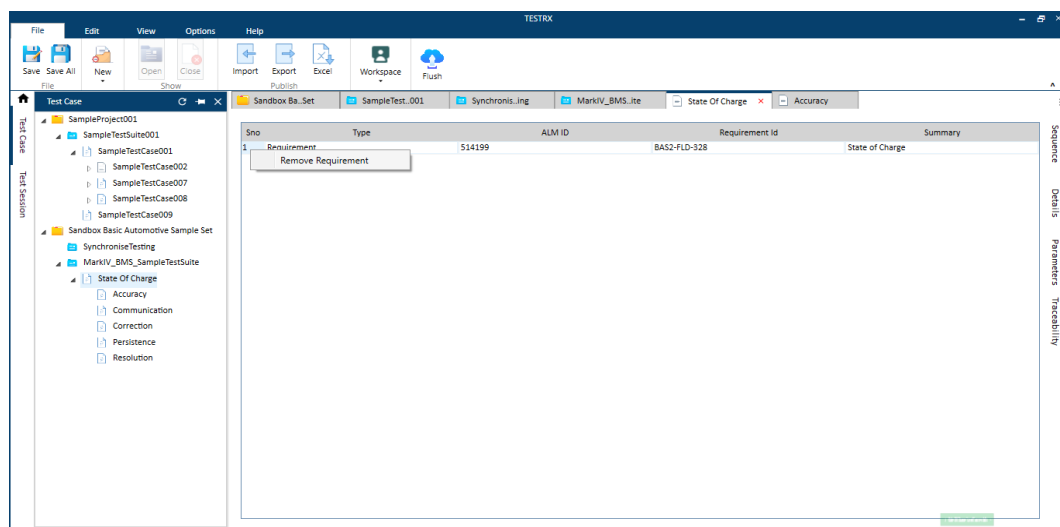
17.1.3 Link Requirements to Test Cases (Traceability)

- To link requirements to a test case, double-click the required test case to open its details view. Navigate to the Traceability tab located on the right side of the screen.
- Within the Traceability table, right-click on the S.No cell and select Add Requirement. Enter the corresponding ALM Requirement ID and press Enter.



Snp.45 – Add Requirement Option

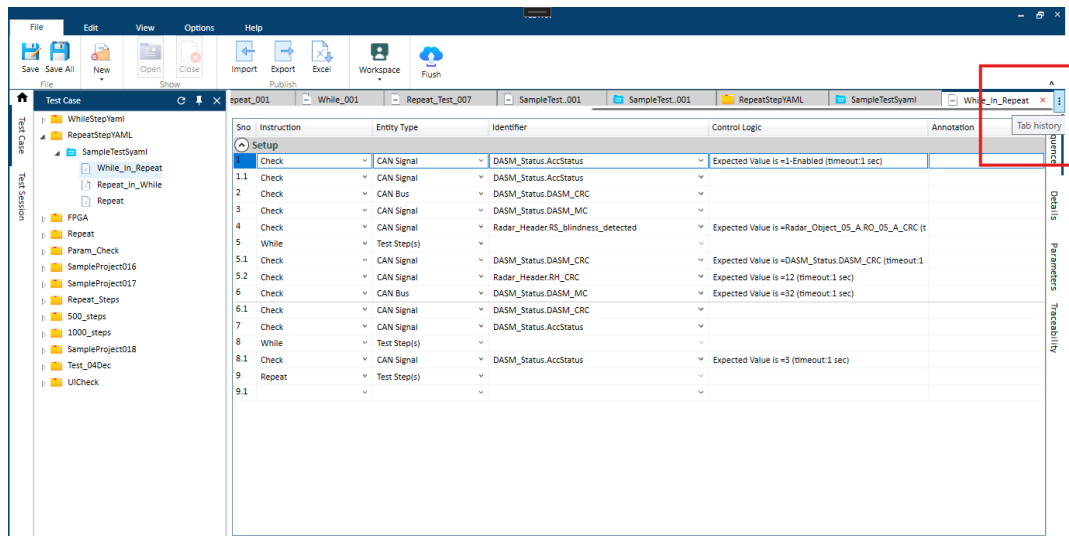
- The requirement ID and summary are automatically fetched from Jama ALM and populated in the table. Multiple requirements can be linked to a single test case as needed.
- To remove a linked requirement, right-click on the S. No cell of the corresponding requirement row and select Remove Requirement. The requirement link is then removed from the test case.



Snp.46 – Remove Requirement Option

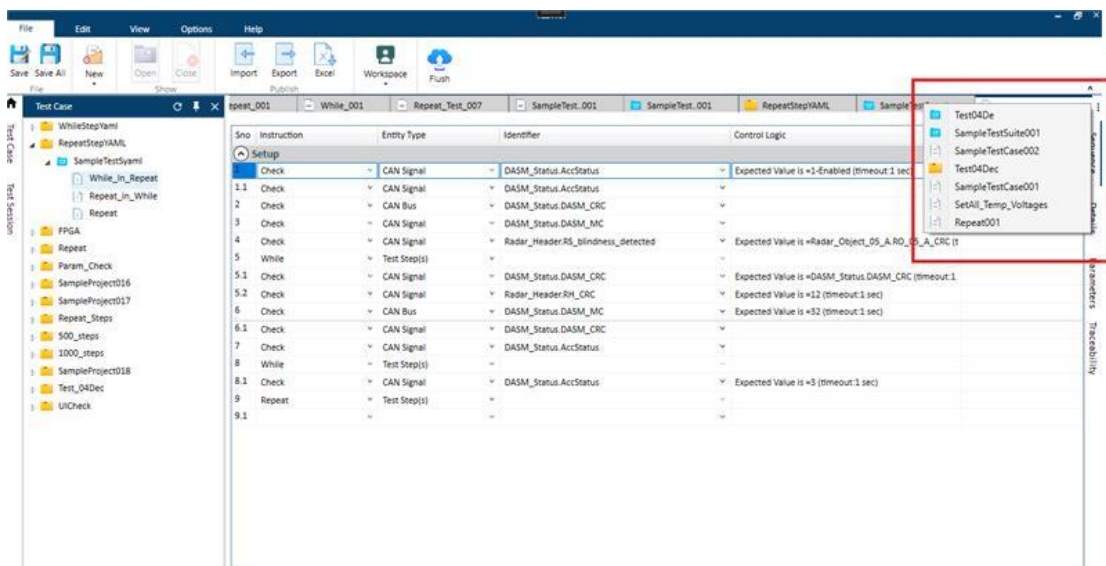
18 Tab Management and Table Options

- When multiple tabs are open and overflow occurs, a **Tabs History** option is available.



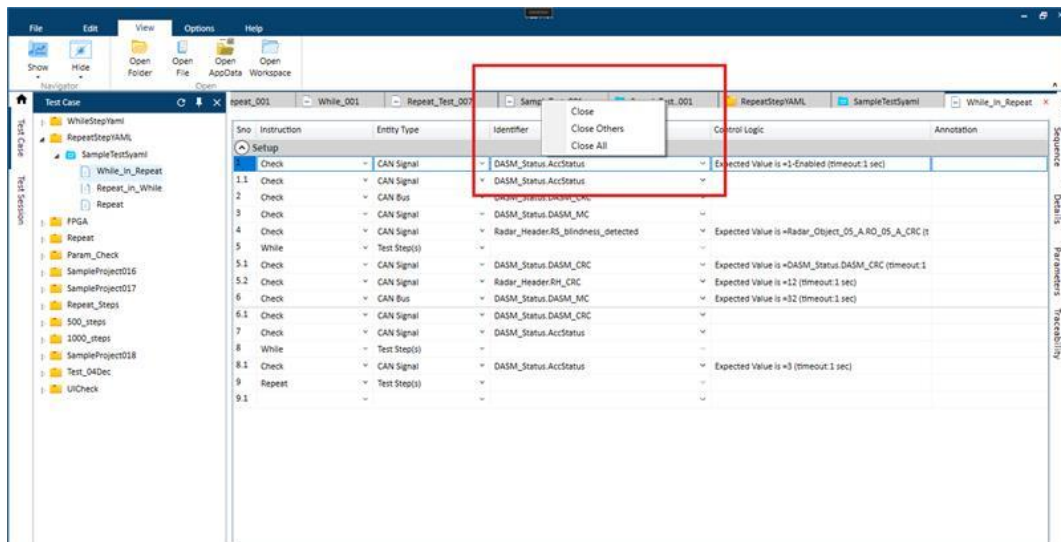
Snp.47 – Tab History Option

- Clicking this option displays a list of overflowed tabs, allowing the user to select and open the required tab.



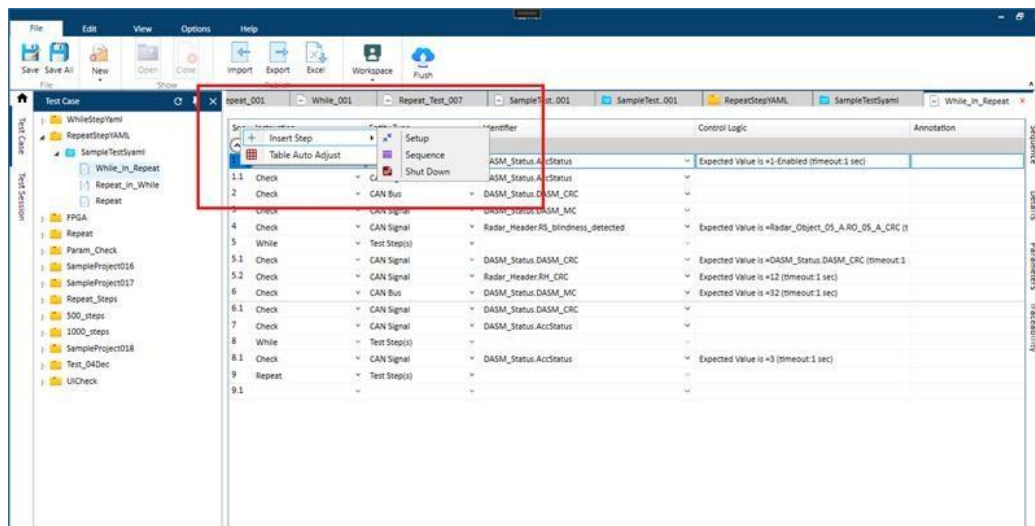
Snp.48 – Overflowed Tabs

- Right-clicking on any tab shows three options: **Close**, **Close Others**, and **Close All**.
 - Close** closes the selected tab.
 - Close Others** closes all tabs except the selected one.
 - Close All** closes all open tabs.



Snp.49 – Tab Management options

- Additionally, if any sections in the Test Case Editor are closed, they can be added back by right-clicking on the **S. No** cell of the test case editor.
- This menu also includes a **Table Auto Adjust** option, which adjusts the table size based on the test case editor content.
- To revert to the default table size, right-click the **S. No** cell again and select **Default Table**.



Snp.50 – Table Management

